A Systematic Benchmark of Persistent Homology Pipelines for Classification

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Abstract

Persistent homology classification pipelines have three stages— filtration, vectorization, and classifier—each with several methods; we benchmarked all three jointly in a controlled factorial sweep of 616 configurations (6 dataset instances, 4 filtrations, 7 vectorizers, 4 classifiers; the 56 configurations pairing the two point-cloud filtrations with image data are excluded as inapplicable — weak Alpha because Delaunay construction in $\mathbb{R}^{28}$ is infeasible, Sparse Rips because its giotto-tda implementation does not complete on the same input). The headline is that the stage ordering is robust while its magnitude is menu-dependent—vectorization stays the dominant stage under a diverse filtration set, per-vectorizer tuning, three persistence libraries, repeated cross-validation, and a regime where topology carries the signal, but one 10-class image dataset (10-class MNIST) inverts the ordering, with filtration leading, while other 10-class datasets remain vectorizer-led (Fashion-MNIST, ModelNet10, Outex), and at matched level counts the gap narrows (ECG200) or reverses (MNIST, ECG5000). On ECG200 the vectorizer marginal range is 6.39pp (95% CI [6.13, 6.65]; $\eta^2 = 0.217$) versus 0.69pp [0.57, 0.81] for filtration; on binary MNIST it is 3.22pp (single split; 3.03pp vs. 1.55pp pooled at r=5) versus 1.65pp, with cubical beating Vietoris–Rips (98.0% vs. 96.25%). ECG5000 shows the menu dependence directly: 24.89pp versus 3.60pp on the raw single-split menu, but 4.30pp versus 0.96pp with disjoint CIs under 25-repetition repeated CV—the ordering persists, the magnitude does not. These results complement, rather than refute, Conti et al. [12]: their 18%–94% range spans a joint filtration-and-vectorization redesign. Noise robustness is not a scale artefact: mean accuracy 99.85% at $\sigma = 0.30$; matched-genus TDA 95.83% (single-split menu; 90.98% under the 25-repetition VR/DTM harmonisation of §4.1) versus 48–58% for norm features; measured bottleneck distances (max 0.434) fall well below the typical-scale bound ($2\sigma\sqrt{2\ln n} \approx 1.82$ at n=100, the Gaussian-max estimate exceeded w.p. ≈93% at d=3; the d=3-corrected value is ≈2.1). The framework is YAML-configurable and extensible to 7 filtrations, 11 vectorizers, and 4 classifiers; all code, configuration, and result schemas are provided.

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Chapter 1

Introduction and Motivation

1.1 Topological Data Analysis and the Pipeline Problem

Topological Data Analysis (TDA) extracts shape invariants from data: connected components, cycles, and voids that persist across multiple scales. Unlike traditional machine learning which operates on raw coordinates or pixel intensities, TDA captures the underlying "shape" of the data — information invariant under stretching and bending. Persistent homology, the workhorse of TDA, tracks the birth and death of topological features as a scale parameter varies.

Practitioners face a combinatorial problem: the TDA classification pipeline has three stages — filtration, vectorization, and classification — each with 5–15 available methods (surveys: Ali et al. [2] and Hensel et al. [18]). The most direct antecedent to "which stage matters most?" comes from Conti et al. [12], who showed that filtration choice on MNIST images can be catastrophic: accuracy ranged from 18% (their H0-only naive arm, a cubical filtration of the negative-greyscale image) to 94% (a redesigned pipeline whose representation is re-selected by grid search), under 10-split cross-validation. That study has sometimes been read as a general principle — "filtration dominates" — although the authors themselves did not claim one.

This paper asks the same question across two real datasets of different modalities — vectorization is the largest single main effect on both ECG200 and binary MNIST (magnitude and, at 10-class scale, ordering treated as menu-dependent in §4.1) — and then generalises the test to a nine-dataset panel, a multi-patient ECG benchmark and a 10-class image probe. The vectorizer–filtration gap is 6.39pp (95% CI [6.13, 6.65]) versus 0.69pp (95% CI [0.57, 0.81]) on ECG200; on binary MNIST the vectorizer range (3.22pp) is twice the filtration range (1.65pp), and cubical beats Vietoris-Rips by ∼1.75pp best-of-family. Filtration effects are dataset-specific and modest on binary MNIST, not the governing stage. We show this using a modular benchmarking frame-

work that varies all three pipeline stages simultaneously in a controlled factorial sweep: 6 dataset instances × 4 filtrations (Vietoris-Rips, weak Alpha, Sparse Rips, Cubical) × 7 vectorizers × 4 classifiers — 672 possible, 616 completed, with the 56 exclusions (`weak_alpha`/`sparse_rips` on MNIST; these point-cloud filtrations are inapplicable to image input). The framework is extensible via YAML-only configuration changes to the wider method families of Appendix D (7 filtrations, 11 vectorizers, and 4 classifiers); standardised harnesses of this kind are the direction the community has moved towards [33].

1.2 The Persistent Homology Pipeline

Whether persistent homology features help classification at all has itself been questioned [34]; Figure 1.1 summarises the three stages this section introduces; this benchmark assumes the features are informative and asks which pipeline stage extracts them most effectively.

A persistent homology classification pipeline transforms raw data through three stages:

1. **Filtration.** Raw data is converted into a nested sequence of simplicial complexes parameterised by a scale ε. Topological features appear (birth at b) and disappear (death at d). Output: a *persistence diagram*, a multiset of (b, d) pairs.

2. **Vectorization.** Diagrams are multisets of variable cardinality and cannot be fed directly into classifiers. Vectorization methods map each diagram to a fixed-length feature vector.

3. **Classification.** The vectorized features are passed to a standard classifier which learns a decision boundary.

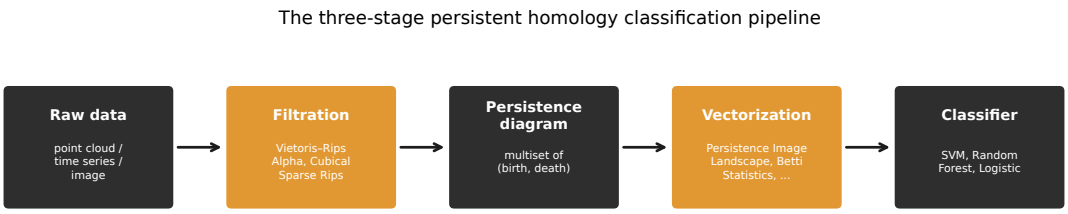


Figure 1.1: The three-stage persistent homology classification pipeline.

1.3 Contributions

This dissertation contributes a reproducible factorial benchmark framework and a set of case studies — two headline datasets, then a nine-dataset panel, a multi-patient ECG benchmark and further probes — that use it to locate the dominant pipeline stage. Specifically:

1. **Framework.** A benchmarking framework that varies all three pipeline stages simultaneously: 4 filtrations, 7 vectorizers, and 4 classifiers executed in a 616-configuration factorial sweep (of 672 possible; the 56 exclusions of §1.1), with YAML-driven configuration and normalised SQLite storage. Via YAML-only changes the framework extends to the full method families of Appendix D (7 filtrations, 11 vectorizers, and 4 classifiers). Full configuration and schema in Appendices A–B; code implementations in Appendix C.

2. **Verified finding: vectorization is the largest main effect.** On ECG200 time series with Takens embedding, vectorization is the largest contributor to classification accuracy. Its 6.39pp marginal range across 7 vectorizers (95% CI [6.13, 6.65]) is roughly nine times the filtration range (0.69pp across 3; 95% CI [0.57, 0.81]) and nearly twice the classifier range (3.50pp across 4; 95% CI [3.28, 3.71]). All three CIs exclude zero, and the ordering vectorizer > classifier > filtration was stable in all 25 cross-validation repetitions. On binary MNIST the vectorizer accounts for most of the accuracy variance (marginal range 3.22pp versus 1.65pp for filtration), and cubical beats Vietoris-Rips by ∼1.75pp best-of-family (98.0% vs. 96.25%); a second time-series dataset (ECG5000) shows the raw-menu replication (24.89pp versus 3.60pp, §4.1); the repeated-CV harmonisation of §4.1 confirms the ordering there (vectorizer 4.30pp versus filtration 0.96pp, disjoint CIs). Filtration choice is second-order on the binary tasks under the Rips-family menus, but not in general: with a non-Rips filtration (DTM) the ECG200 filtration range grows to 2.81pp and DTM is the best filtration, winning 7 of 8 vectorizer–classifier pairs (§4.2); the 10-class inversion of §4.1 and the population-level mixed model of §4.4 both rank filtration above vectorization in their settings. Conti et al.'s MNIST case study (18% with their cubical pipeline to 94% with improved filtrations, 10-class, grid search) is a complementary demonstration that a badly-matched filtration can be catastrophic on images; it does not establish a general law.

3. **Matched-scale synthetic evidence.** On synthetic genus-1 versus genus-2 point clouds whose radial norm distributions are quantile-matched, norm features classify at chance (48–58%) while the TDA pipeline sustains 95.83% accuracy at

$\sigma = 0.30$ noise — the noise robustness survives the norm/scale confound between the sphere and torus geometries.

4. **Stability-bound confirmation.** Measured bottleneck distances between clean and noisy diagrams (max $d_B = 0.434$ over 80 cloud/dimension pairs) sit well below the corrected worst-case bound $d_B \leq 2\sigma\sqrt{2 \ln n}$ (≈ 1.82 at $\sigma = 0.30$, $n = 100$), confirming the bound is conservative in practice.

Chapter 2 develops the mathematical machinery needed to understand why these pipeline stages interact as they do.

Chapter 2

Mathematical Foundations of Persistent Topology

This chapter develops the algebraic and geometric machinery underlying persistent homology. We progress from simplicial complexes through chain groups, homology vector spaces, filtrations, and persistence modules, concluding with the stability theorem and vectorization methods that convert diagrams into classifier-ready feature vectors.

2.1 Simplicial Complexes

Definition 2.1 (Simplex)**.** A *k-simplex* $\sigma = [v_0, v_1, \ldots, v_k]$ is the convex hull of $k+1$ affinely independent points in $\mathbb{R}^d$ ($d \geq k$). A 0-simplex is a vertex, a 1-simplex an edge, a 2-simplex a triangle, a 3-simplex a tetrahedron. A *face* of σ is the convex hull of any non-empty subset of its vertices.

Definition 2.2 (Simplicial Complex)**.** A *simplicial complex* K is a finite collection of simplices such that:

1. If $\sigma \in K$ and τ is a face of σ, then $\tau \in K$ (closure under taking faces).
2. If $\sigma, \tau \in K$, then $\sigma \cap \tau$ is either empty or a face of both.

The *k-skeleton* $K^{(k)}$ is the subcomplex of all simplices of dimension at most k. The 1-skeleton is a graph in the usual sense.

2.2 Filtrations and Complex Constructions

Definition 2.3 (Filtration)**.** A *filtration* of a simplicial complex K is a nested sequence of subcomplexes indexed by a real parameter $\varepsilon \geq 0$:

$$\emptyset = K_{\varepsilon_0} \subseteq K_{\varepsilon_1} \subseteq \cdots \subseteq K_{\varepsilon_m} = K, \tag{2.1}$$

where $\varepsilon_0 < \varepsilon_1 < \cdots < \varepsilon_m$.

Three filtrations are used in the benchmark:

Vietoris-Rips complex $\mathrm{VR}_\varepsilon(X)$. For a finite point cloud $X \subset \mathbb{R}^d$, a k-simplex $[x_0, \ldots, x_k]$ is included if $\|x_i - x_j\| \leq \varepsilon$ for all $0 \leq i < j \leq k$. The Rips complex is simple to compute but large: for n points, the number of k-simplices scales as $O(n^{k+1})$. Specifically, the number of 2-simplices (triangles) in the full Rips complex is $O(n^3)$, which drives the runtime differences observed in Chapter 4.

Alpha complex $A_\varepsilon(X)$. Constructed via the intersection of Voronoi cells with metric balls. For each point $x_i \in X$, let $V_i = \{y \in \mathbb{R}^d : \|y - x_i\| \leq \|y - x_j\| \text{ for all } j \neq i\}$ be its closed Voronoi cell, and define $R_i(\varepsilon) = V_i \cap \overline{B}_{\varepsilon/2}(x_i)$ where $\overline{B}_{\varepsilon/2}(x_i)$ is the closed ball of radius $\varepsilon/2$. The Alpha complex is the nerve of the cover $\{R_i(\varepsilon)\}_{i=1}^n$. The Nerve Theorem (see below) guarantees homotopy equivalence to the union of balls.

Theorem 2.1 (Nerve Theorem — Leray 1945 [22])**.** *Let $\mathcal{U} = \{U_i\}_{i \in I}$ be a finite open cover of a topological space M such that every non-empty intersection $\bigcap_{j \in J} U_j$ is contractible. Then the nerve $N(\mathcal{U})$ is homotopy equivalent to $\bigcup_{i \in I} U_i$.*

For the Alpha cover $\{R_i(\varepsilon)\}$, the sets $R_i(\varepsilon)$ are convex polyhedra in $\mathbb{R}^d$, so all finite intersections are convex and thus contractible. The Nerve Theorem therefore guarantees that $A_\varepsilon(X) \simeq \bigcup_i \overline{B}_{\varepsilon/2}(x_i)$. In $\mathbb{R}^3$, the Alpha complex is a subcomplex of the Delaunay triangulation with $O(n^2)$ simplices, compared to $O(n^3)$ for the full Rips complex.

Remark 2.1. The benchmark executes giotto-tda's `WeakAlphaPersistence`, which per giotto's documentation is the Vietoris–Rips filtration of the sparse matrix of distances between Delaunay-neighbouring vertices — a subcomplex of the full Rips complex (cheaper to build), not a subcomplex of the Alpha complex, and not covered by the Nerve-Theorem homotopy guarantee, which applies to the full Alpha/Čech complex. For the executed approximation we verify empirically in §4.5 that the full GUDHI Alpha complex (persistence-equivalent to the Čech complex) reproduces weak-alpha classification accuracy (mean difference 0.0–0.8pp, measured in a dedicated GUDHI-versus-giotto run, scripts/sweeps/experiment_alpha.py) at 2–5× the runtime cost, so the approximation is faithful on these tasks.

Sparse Rips complex. A subsampled approximation of VR that retains only a sparse subset of edges and higher simplices, designed for point clouds with $n \geq 10^3$ points where full VR computation is prohibitive. The sparsification parameter ϵ controls the approximation fidelity.

2.3 Homology Groups and Persistent Homology

Chain Complexes and Homology

To define persistent homology formally, we first construct the algebraic machinery of simplicial homology over the field $\mathbb{Z}_2 = \mathbb{Z}/2\mathbb{Z}$.

Definition 2.4 (Chain Group)**.** For a simplicial complex K, the *k-th chain group* $C_k(K; \mathbb{Z}_2)$ is the $\mathbb{Z}_2$-vector space whose basis elements are the k-simplices of K. An element $c \in C_k(K; \mathbb{Z}_2)$ is a formal sum $c = \sum_{\sigma \in K^{(k)}} a_\sigma \sigma$ with coefficients $a_\sigma \in \mathbb{Z}_2$.

Definition 2.5 (Boundary Operator)**.** The *k-th boundary operator* $\partial_k : C_k(K; \mathbb{Z}_2) \to C_{k-1}(K; \mathbb{Z}_2)$ is the linear map defined on a k-simplex $\sigma = [v_0, \ldots, v_k]$ by:

$$\partial_k [v_0, \ldots, v_k] = \sum_{i=0}^{k} [v_0, \ldots, \hat{v}_i, \ldots, v_k], \tag{2.2}$$

where $\hat{v}_i$ denotes omission of v_i, and the sum is taken over $\mathbb{Z}_2$ (so $1+1=0$). For $k = 0$, $\partial_0 = 0$ by convention.

The fundamental property of the boundary operator is $\partial_k \circ \partial_{k+1} = 0$ for all $k \geq 0$. Intuitively, the boundary of a boundary is empty: the edges bounding a triangle form a closed loop with no endpoints.

Definition 2.6 (Homology Group)**.** For a simplicial complex K, define:

- $Z_k(K) = \ker \partial_k$ — the *k-cycles* (chains with zero boundary).
- $B_k(K) = \operatorname{im} \partial_{k+1}$ — the *k-boundaries* (chains that are boundaries of $(k+1)$-chains).

Since $\partial_k \circ \partial_{k+1} = 0$, we have $B_k(K) \subseteq Z_k(K)$. The *k-th homology group* is the quotient vector space:

$$H_k(K) = \frac{Z_k(K)}{B_k(K)} = \frac{\ker \partial_k}{\operatorname{im} \partial_{k+1}}. \tag{2.3}$$

Elements of $H_k(K)$ are equivalence classes of k-cycles, where two cycles are equivalent if they differ by a boundary. The dimension of $H_k(K)$ as a $\mathbb{Z}_2$-vector space is the *k-th Betti number*: $\beta_k(K) = \dim_{\mathbb{Z}_2} H_k(K)$. Informally, β_0 counts connected components, β_1 counts 1-dimensional holes (cycles), β_2 counts 2-dimensional voids.

Persistent Homology

A filtration $\{K_\varepsilon\}_{\varepsilon \geq 0}$ induces a sequence of inclusion maps $i^{\varepsilon,\varepsilon'} : K_\varepsilon \hookrightarrow K_{\varepsilon'}$ for $\varepsilon \leq \varepsilon'$, which in turn induce linear maps on homology:

$$f_k^{\varepsilon,\varepsilon'} : H_k(K_\varepsilon) \to H_k(K_{\varepsilon'}). \tag{2.4}$$

Definition 2.7 (Persistent Homology Group)**.** The *k-th persistent homology group* from ε to ε' is:

$$H_k^{\varepsilon,\varepsilon'}(K) = \operatorname{im} f_k^{\varepsilon,\varepsilon'}. \tag{2.5}$$

A homology class $\gamma \in H_k(K_{\mathrm{b}})$ is *born* at $\varepsilon = \mathrm{b}$ if $\gamma \notin \operatorname{im} f_k^{\mathrm{b}-\delta,\mathrm{b}}$ for any $\delta > 0$, and *dies* at $\varepsilon = \mathrm{d} > \mathrm{b}$ if $f_k^{\mathrm{b},\mathrm{d}}(\gamma) \in \operatorname{im} f_k^{\mathrm{b}-\delta,\mathrm{d}}$ for some $\delta > 0$ but $f_k^{\mathrm{b},\mathrm{d}-\delta}(\gamma) \notin \operatorname{im} f_k^{\mathrm{b}-\delta,\mathrm{d}-\delta}$. If γ never dies, we set $\mathrm{d} = \infty$.

Definition 2.8 (Persistence Diagram)**.** The *persistence diagram* $\mathrm{Dgm}_k(K)$ is the multiset of points $(\mathrm{b},\mathrm{d}) \in \mathbb{R}^2 \cup \{\infty\}$ where $\mathrm{b} < \mathrm{d}$, together with infinitely many copies of each point on the diagonal $\Delta = \{(x,x) : x \geq 0\}$. The *persistence* of a feature is $\mathrm{p} = \mathrm{d} - \mathrm{b}$.

Example. A sphere S^2 has $\beta = (1,0,1)$: one component, no 1-cycles, one 2-void. A torus T^2 has $\beta = (1,2,1)$: one component, two independent 1-cycles (the meridional and longitudinal loops), one 2-void. This $\beta_1 = 0$ versus $\beta_1 = 2$ difference is the topological signal driving the sphere-versus-torus classification in Chapter 4.

2.4 Stability of Persistence Diagrams

The stability theorem guarantees that small perturbations of the input data produce proportionally small changes in the persistence diagram — essential for any pipeline handling noisy data.

Definition 2.9 (Bottleneck Distance)**.** The *bottleneck distance* between two persistence diagrams D_1 and D_2 is:

$$d_B(D_1, D_2) = \inf_{\gamma : D_1 \to D_2} \sup_{p \in D_1} \|p - \gamma(p)\|_\infty, \tag{2.6}$$

where γ ranges over all bijections from D_1 to D_2 (including pairings with the diagonal Δ), and $\|\cdot\|_\infty$ is the L^∞ norm on $\mathbb{R}^2$.

Definition 2.10 (Gromov-Hausdorff Distance)**.** For compact metric spaces X and Y, the *Gromov-Hausdorff distance* is:

$$d_{GH}(X,Y) = \frac{1}{2} \inf_{Z,f,g} d_H^Z(f(X), g(Y)), \tag{2.7}$$

where the infimum is taken over all metric spaces Z and all isometric embeddings $f : X \hookrightarrow Z$, $g : Y \hookrightarrow Z$, and d_H^Z is the Hausdorff distance in Z: $d_H^Z(A,B) = \max\{\sup_{a\in A} \inf_{b\in B} d_Z(a,b),\ \sup_{b\in B} \inf_{a\in A} d_Z(a,b)\}$.

Theorem 2.2 (Stability Theorem for geometric complexes — Chazal, de Silva & Oudot [8])**.** *Let X, Y be totally bounded metric spaces. Then:*

$$d_B(\mathrm{Dgm}_k(X), \mathrm{Dgm}_k(Y)) \leq 2\, d_{GH}(X, Y) \tag{2.8}$$

for all $k \geq 0$, where $\mathrm{Dgm}_k(\cdot)$ denotes the k-dimensional persistence diagram of the Vietoris-Rips or Čech filtration (Chazal, de Silva & Oudot 2014, Thm. 5.2). The function-level stability bound $d_B \leq \|f - g\|_\infty$ of Cohen-Steiner, Edelsbrunner & Harer [11] applies instead to the sublevel-set (cubical) filtration on grids.

This theorem is applied in §4.2: for n points with additive Gaussian noise $\eta_i \sim \mathcal{N}(0, \sigma^2 I_3)$, extreme value theory gives the typical scale of the maximum perturbation as $\mathbb{E}[\max_i \|\eta_i\|] \approx \sigma\sqrt{2 \ln n}$ (the one-dimensional leading-order value; for the executed $d = 3$ per-coordinate noise the expectation is larger by the dimension correction, $\approx 1.16\,\sigma\sqrt{2 \ln n}$ at $n = 100$; Monte Carlo gives $\mathbb{E}[\max_i \|\eta_i\|] \approx 1.06$ at $\sigma = 0.30$). The noisy cloud therefore lies within Gromov–Hausdorff distance of order $\sigma\sqrt{\ln n}$ of the clean one — a typical-case estimate, not an almost-sure bound: the threshold $\sigma\sqrt{2 \ln n}$ is exceeded by the maximum with probability $\approx 93\%$ in the executed d=3 noise (the 1-D value would be $\approx 22\%$). By Theorem 2.2 the bottleneck distance between clean and noisy diagrams satisfies

$$d_B\Big(\mathrm{Dgm}_k(X),\, \mathrm{Dgm}_k(X + \eta)\Big) \;\lesssim\; 2\,\sigma\sqrt{2 \ln n}. \tag{2.9}$$

The factor 2 inherited from the stability theorem is essential: for $n = 100$ (points per cloud after subsampling), $2\,\sigma\sqrt{2 \ln 100} \approx 0.91$ at $\sigma = 0.15$ and ≈ 1.82 at $\sigma = 0.30$ — a factor of two larger than the bound one would write down by omitting it. Measured bottleneck distances between clean and noisy diagrams (ripser diagrams [5], GUDHI bottleneck implementation [23]; 80 cloud/dimension pairs; §4.2) are far below this: at $\sigma = 0.15$ the mean is 0.144 and the maximum 0.300; at $\sigma = 0.30$ the maximum is 0.434. The torus H_1 diagrams move by d_B on average 0.283 at $\sigma = 0.30$ — below the measured H_1 persistence range [0.556, 1.156] — so the features comfortably survive the perturbation.

2.5 Vectorization Methods

Vectorization converts a persistence diagram into a fixed-length feature vector for classification. We use the taxonomy of Ali et al. [2]; comparative studies include Barnes et al. [4], Perea et al. [27], and Sulowska [30].

Persistence diagrams are multisets of variable cardinality. To use them with standard classifiers, we must map each diagram to a fixed-length vector in a normed space.

We present the seven methods executed in the sweep (Figure 2.1 shows the four classical representations applied to one diagram): the four classical representations below, followed by three widely used alternatives (Silhouette, Amplitude, and Persistence Entropy).

Persistence Statistics. For each homology dimension k, compute the mean, standard deviation, minimum, and maximum of births, deaths, and lifespans ($\mathrm{p} = \mathrm{d} - \mathrm{b}$) across all points in Dgm_k. This yields $4 \times 3 = 12$ features per dimension. Despite zero hyperparameters, this method is competitive with kernel-based approaches [2].

Betti Curves [10]. The Betti curve $\beta_k(\varepsilon)$ counts the number of k-dimensional features with $\mathrm{b} \leq \varepsilon < \mathrm{d}$. Evaluated at n_{bins} equally spaced points in $[\varepsilon_{\min}, \varepsilon_{\max}]$, this produces n_{bins} features per homology dimension.

Persistence Landscapes [6]. For each point $(\mathrm{b}, \mathrm{d}) \in \mathrm{Dgm}_k$, define the tent function:

$$\Lambda_{(\mathrm{b},\mathrm{d})}(t) = \max\Big(0,\, \min(t - \mathrm{b},\, \mathrm{d} - t)\Big). \tag{2.10}$$

The *m-th persistence landscape* is the function $\lambda_m : \mathbb{R} \to \mathbb{R}$ given by:

$$\lambda_m(t) = m\text{-}\max_{p \in \mathrm{Dgm}_k} \Lambda_p(t), \tag{2.11}$$

where m-max denotes the m-th largest value. Each λ_m is 1-Lipschitz. The sequence $(\lambda_m)_{m \in \mathbb{N}}$ resides in the Banach space $L^p(\mathbb{N} \times \mathbb{R})$ for any $1 \leq p \leq \infty$, enabling classical statistical operations (means, variances, confidence intervals) directly on the vectorized representation. With n_{layers} layers evaluated at n_{bins} points, this produces $n_{\text{layers}} \times n_{\text{bins}}$ features.

Persistence Images [1]. Transform each point $(\mathrm{b}, \mathrm{d}) \in \mathrm{Dgm}_k$ to birth-persistence coordinates (b, p) where $\mathrm{p} = \mathrm{d} - \mathrm{b}$. The weighting may be any non-negative function $w : \mathbb{R} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$; a common choice is a linear ramp $w(\mathrm{b}, \mathrm{p}) = \mathrm{p}/\mathrm{p}_{\max}$ capped at 1. In this study the executed configuration passes *no* weight function to giotto-tda's `PersistenceImage` (Appendix A lists `sigma: 0.1, n_bins: 20` only), and the library's default is constant weights ($w \equiv 1$; verified in giotto-tda 0.6.2, where `weight_function=None` maps to `numpy.ones_like`), so the executed persistence images are *unweighted*. The persistence surface is:

$$\rho(x, y) = \sum_{(\mathrm{b},\mathrm{p}) \in T(\mathrm{Dgm}_k)} w(\mathrm{b}, \mathrm{p})\, \phi_{(\mathrm{b},\mathrm{p})}(x, y), \tag{2.12}$$

where $\phi_{(\mathrm{b},\mathrm{p})}(x,y) = \frac{1}{2\pi\sigma_{\mathrm{PI}}^2}\exp\!\Big(-\frac{(x-\mathrm{b})^2+(y-\mathrm{p})^2}{2\sigma_{\mathrm{PI}}^2}\Big)$ (the kernel bandwidth σ_{PI} is a smoothing hyperparameter, distinct from the noise level σ of §3.2) is a 2D Gaussian kernel. Evaluating ρ on an $n_{\mathrm{bins}} \times n_{\mathrm{bins}}$ grid yields n_{bins}^2 features per homology dimension.

Silhouette [9]. The silhouette of a diagram weights the landscape tent functions of (2.10) by persistence and normalises:

$$\kappa_{\mathrm{Dgm}_k}(t) = \frac{\sum_{p\in\mathrm{Dgm}_k} w(p)\,\Lambda_p(t)}{\sum_{p\in\mathrm{Dgm}_k} w(p)}, \qquad w(p) = \mathrm{p}(p)^q, \tag{2.13}$$

with $q = 1$ in our configuration; evaluating ϕ at n_{bins} points yields n_{bins} features per homology dimension. Silhouette is the top-performing vectorizer on ECG200 (§4.1).

Amplitude . The amplitude of a diagram is its distance to the empty diagram in a diagram metric:

$$A(\mathrm{Dgm}_k) = d\Big(\mathrm{Dgm}_k, \Delta\Big), \tag{2.14}$$

with the bottleneck distance of (2.6) in our configuration — one scalar per homology dimension and the fastest vectorizer in the sweep. On ECG200 it is among the weaker representations (§4.1), indicating that representation capacity is dataset-dependent.

Persistence Entropy [28]. For a diagram with persistences $\mathrm{p}_i = \mathrm{d}_i - \mathrm{b}_i$, normalise $q_i = \mathrm{p}_i / \sum_j \mathrm{p}_j$ and define

$$E(\mathrm{Dgm}_k) = -\sum_i q_i \log q_i, \tag{2.15}$$

one scalar per homology dimension measuring how concentrated the persistence mass is. Entropy is the weakest vectorizer on ECG200 (§4.1): information content is not monotone in representational simplicity.

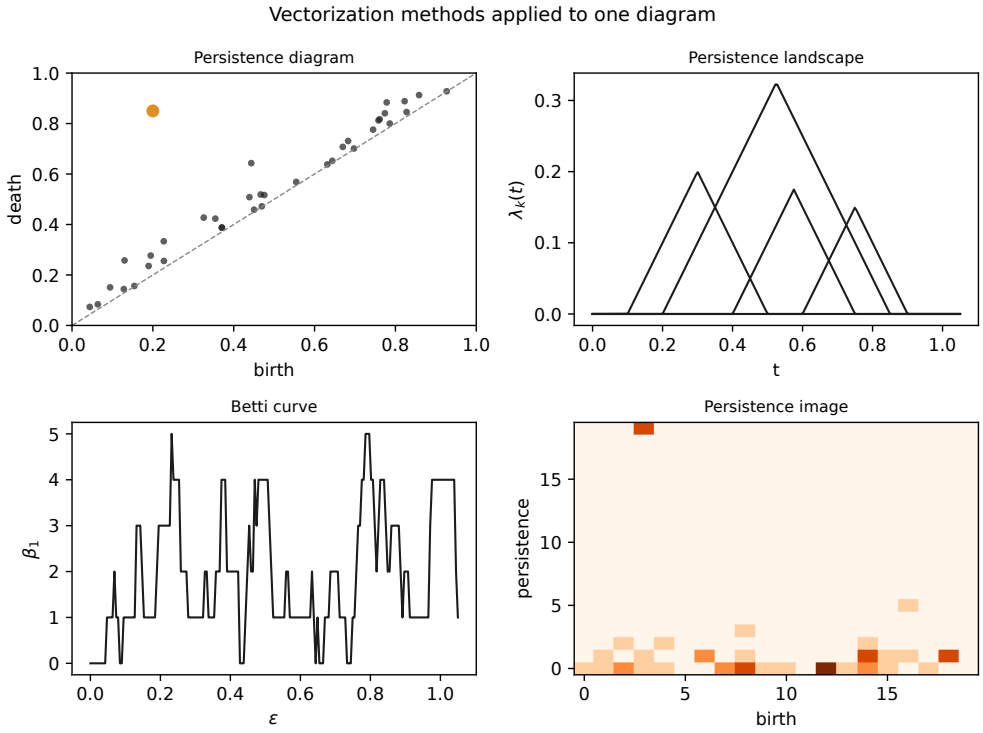


Figure 2.1: The four classical vectorization methods applied to one diagram.

2.6 Mapping Theory to Benchmark Variables

The preceding definitions directly parameterise the benchmark executed in Chapter 4:

- **Simplicial complex size** (§2.1–§2.2): VR's $O(n^3)$ 2-simplex count versus weak Alpha's $O(n^2)$ drives the 3–27% wall-clock speed gap measured in §4.5. The Nerve Theorem guarantees that the full Alpha complex's speed advantage does not cost geometric fidelity; the executed `WeakAlphaPersistence` is an approximation of that complex, and §4.3 verifies it matches full-complex accuracy.

- **Filtration choice** (§2.2): VR, weak Alpha, Sparse Rips, and Cubical constitute the four filtrations varied in the factorial sweep of §4.1.

- **Persistence and stability** (§2.3–§2.4): The corrected bound of (2.9), $d_B \lesssim 2\sigma\sqrt{2\ln n}$, predicts that $\sigma = 0.15$ noise perturbs diagrams by $\lesssim 0.91$ (and $\sigma = 0.30$ by $\lesssim 1.82$), while torus β_1 features have measured top-2 persistence range [0.556, 1.156] (ripser on the shipped clouds). §4.2 confirms that classification survives this perturbation (measured d_B: max 0.300 at $\sigma = 0.15$, max 0.434 at $\sigma = 0.30$).

- **Vectorization** (§2.5): Persistence Statistics, Betti Curves, Landscapes, Images, Silhouette, Amplitude, and Persistence Entropy constitute the seven vectorizers

varied in §4.1, spanning the complexity spectrum from zero-parameter to kernel-based.

- **Classifiers** (§3.1): Logistic Regression (linear baseline), SVM-RBF, SVM-linear, and Random Forest constitute the four classifiers in the sweep. §4.1 tests whether non-linear classifiers add value after vectorization.

Chapter 3

Algorithmic Pipeline and Benchmarking Framework

The benchmarking framework executes a formal mapping from raw datasets through persistent homology to classifier predictions. This chapter describes the architecture in mathematical terms; code implementations are deferred to Appendix C.

3.1 Pipeline Mappings

The full pipeline is the composition of four transformations:

$$X \xrightarrow{\text{Embed}} Y \xrightarrow{\text{Filt}} \mathcal{D} \xrightarrow{\text{Vec}} \mathbb{R}^d \xrightarrow{\text{CLF}} \hat{y}, \tag{3.1}$$

where:

- **Embed** (optional, time series only). Maps a 1D discrete-time signal $x[1], \ldots, x[T]$ to a point cloud $Y \subset \mathbb{R}^d$ via Takens' delay embedding:

$$\Phi_{(d,\tau)}(x)_t = \Big(x[t], x[t+\tau], \ldots, x[t+(d-1)\tau]\Big), \quad t = 1, \ldots, T-(d-1)\tau. \tag{3.2}$$

  By Takens' Delay Embedding Theorem [31], if the signal x is a generic smooth measurement of a dynamical system on a compact manifold M of dimension d_M, then $\Phi_{(d,\tau)}$ is an embedding provided $d > 2d_M$. We use the pragmatic defaults $d = 3$, $\tau = 1$ throughout. Delay-embedding TDA is standard practice for time-series classification [26, 35]; the recorded signals are not established as generic measurements of a known-dimension attractor, so the theorem's $d > 2d_M$ condition is not asserted. Embedding-parameter selection criteria are discussed by Kennel et al. [20] and Fraser–Swinney [16]. Sensitivity to (d, τ) is examined in Appendix D.

- **Filt** (§2.2). Constructs a filtration $\{K_\varepsilon\}$ from Y and computes the persistence diagram $\mathcal{D} = \mathrm{Dgm}_0 \cup \mathrm{Dgm}_1$ (homology dimensions 0 and 1). Implemented via giotto-tda's VietorisRipsPersistence, WeakAlphaPersistence, SparseRipsPersistence, or CubicalPersistence transformers.

- **Vec**. Maps $\mathcal{D}$ to a vector $v \in \mathbb{R}^d$ via one of the seven methods in §2.5. Global grid bounds ($\mathrm{b}_{\min}, \mathrm{b}_{\max}, \mathrm{p}_{\max}$) are computed on the training fold only and applied to the test fold without re-fitting, preventing cross-validation data leakage.

- **CLF**. A scikit-learn classifier (LogisticRegression, SVC with linear or RBF kernel, or RandomForestClassifier with 100 trees).

3.2 Point Cloud Preprocessing

Subsampling. Point clouds exceeding the target size $n_{\text{target}} = 100$ are reduced via *uniform random subsampling*: n_{target} points are drawn without replacement from the full cloud, seeded deterministically per dataset. Uniform random sampling does not preserve metric-space geometry as well as farthest-point sampling would; we use it for simplicity and reproducibility. We tested whether the choice matters: greedy farthest-point sampling (FPS) was compared against this uniform-random scheme on the synthetic sphere/torus clouds, reduced to $k \in \{50, 15\}$ points (the native clouds carry 100 points, so the runner's `subsample_points` knob never fires and the subsampling choice must be made below the native resolution to be nontrivial), under both noise levels. The grid spanned {`vietoris_rips`, `weighted_rips`(DTM)}× {`betti_curve`, `persistence_landscape`}× {`random_forest`, `svm_rbf`} (8 configurations per arm, stratified 5-fold CV, seed 42, rep 1; `fps_ablation.db`).

At the design resolution (k=50) both methods attain 100.00% mean accuracy at both σ=0 and σ=0.30 — the task saturates and the two arms are indistinguishable. In the only regime where a difference emerges (k=15), FPS is *not* better: at σ=0 FPS scores 99.88% vs. uniform 100.00% (−0.12 pp), and at σ=0.30 FPS scores 98.94% vs. uniform 99.81% (−0.88 pp). Across all 64 runs the minimum configurational accuracy is 98.5%, even after a 7× reduction in point count. Uniform-random subsampling is not materially inferior to FPS on these point clouds — if anything it is marginally *better* once the task is pushed below saturation.

Because the sphere/torus problem remains separable under even aggressive subsampling, it does not expose a regime where sampling density matters; a dataset with fragile or heterogeneously sampled topology would be needed to observe a gap. The subsampling-method choice is immaterial for the synthetic class, where both sampling schemes saturate the task (the FPS ablation of §4.2 confirms no accuracy gap at the benchmark's sampling density).

Noise model. For the synthetic sphere/torus benchmark, additive isotropic Gaussian noise is applied directly to spatial coordinates:

$$x_i \mapsto x_i + \eta_i, \quad \eta_i \sim \mathcal{N}(0, \sigma^2 I_3), \tag{3.3}$$

where I_3 is the 3×3 identity matrix. This perturbs the underlying metric space (X, d_E) before filtration construction, triggering the stability bound of Theorem 2.2. (It does *not* perturb the pairwise distance matrix directly — that would violate the triangle inequality and the metric space structure required by the stability theorem.)

3.3 Software Architecture

The framework uses **giotto-tda 0.6.2** [32] for all filtrations and vectorizations (scikit-learn-compatible transformers), **scikit-learn 1.3.2** for classifiers and cross-validation, with **Ripser 0.6.15** [5] and **GUDHI 3.13.0** [23] as alternative backends. Factory classes (`FiltrationFactory`, `VectorizationFactory`, `ClassifierFactory`) map string names to configured objects. A YAML file (Appendix A) defines the sweep space declaratively. Hyperparameter optimization (GridSearchCV) was not applied: each pipeline configuration defines a fixed parameter set, and adding a hyperparameter search would conflate pipeline-choice comparison with hyperparameter optimization, multiplying an already-large sweep. The `PipelineRunner` iterates the Cartesian product of all choices, runs stratified 5-fold cross-validation, and stores per-fold metrics (accuracy, F1, precision, recall) in a normalised SQLite database (Appendix B) with a config snapshot for reproducibility (retained for the main sweep; the additional-sweep DBs — panel, hyperparameter, large-n, MNIST-10 — do not retain one; their provenance is the driver scripts in scripts/). Reported 95% CIs are computed from repeated cross-validation (repeated stratified 5-fold CV with independent splits: 25 repetitions, seeds 43–67, for the headline ECG200 analysis; 5 repetitions, seeds 43–47, for MNIST): per-stage marginal ranges are computed within each repetition, and the CI is $\bar{x} \pm t_{0.025, n_{\text{reps}}-1} \cdot s/\sqrt{n_{\text{reps}}}$ over the repetition-level estimates, which captures the between-split variance directly. Per-configuration intervals use the Nadeau–Bengio corrected resampled estimator $\text{SE}^2 = (1/(kr) + n_2/n_1)\, s^2$ over the kr fold accuracies ($kr = 125$ for the ECG200 r=25 analysis, 25 for MNIST). Per-configuration accuracy varies across repetitions with mean SD 1.09pp (max 3.11pp, r=5), so single-split point estimates are noisy. Stage dominance is quantified primarily by an η^2 variance decomposition — a main-effects-only three-way ANOVA over per-config means and over fold-level accuracies, with F-tests (Table 4.3 in §4.1). No paired Wilcoxon tests are used: CV folds are not independent replicates, and with $n = 5$ pairs the smallest achievable two-sided p-value is 0.0625. Adding a new filtration, vectorizer, or dataset

requires editing only the YAML file; adding a fundamentally new component type (e.g., learned vectorizers) requires new factory code.

3.4 Data and Code Availability

Data and code availability. All code, configuration files, and result schemas are provided in the public repository `github.com/KRUZZZZY/tda-benchmark` (MIT licence). The exact build described here is tagged `v1.0.1-dissertation` and archived, together with its result databases and flat CSV exports, under DOI `10.5281/zenodo.XXXXXXX` (to be inserted after archiving); `REPRODUCING.md` maps every result to its producing script, database, and runtime, and the data-fetch scripts verify SHA256 checksums. The synthetic sphere/torus data are regenerated by the bundled `scripts/data/generate_datasets.py` (seeded; 200 samples, 100 points per cloud, noise baked into coordinates); ECG200 is the UCR archive benchmark [13]; MNIST is the standard binary 0/1 subset [21]. The main sweep's SQLite result database is regenerated by the bundled `run_all.sh` script, and every follow-up analysis has a bundled producer: `scripts/sweeps/sweep_repeated_cv_r25.py` writes `data/tda/repeated_cv_r25.db` (2100 finished runs: 84 configurations $\times$ 25 repetitions); `scripts/sweeps/run_baselines.py` writes `data/tda/baseline_experiments.db` (raw-signal baselines, matched-genus synthetic clouds, measured bottleneck distances); `scripts/data/generate_matched_synthetic.py` writes the `data/tda/synthetic_matched/` data; and `scripts/analysis/analysis_repeated_cv_r25.py`, `scripts/analysis/analysis_eta_omega2.py`, `scripts/analysis/analysis_mnist_repeated_cv.py`, `scripts/sweeps/experiment_alpha.py`, `scripts/sweeps/experiment_cubical_artifact.py`, and `scripts/sweeps/ecg5000_lean_sweep.py` produce the remaining analyses and sensitivity results. REPRODUCING.md maps each database to its producer. Random seeds are fixed: the main sweep uses base seed 42, the repeated-CV splits use seeds 43–67 (random seed 42 + repetition index; 25 repetitions for ECG200, 5 for MNIST), and per-dataset subsampling is seeded deterministically via CRC32, so all experiments are reproducible end-to-end. Chapter 4 reports the empirical results this architecture produces.

Chapter 4

Empirical Benchmark and Sensitivity Analysis

One benchmark sweep, four analytical cuts. Table 4.1 lists the instances. The sweep comprises 616 completed configurations from a full factorial of 6 dataset instances × 4 filtrations (Vietoris-Rips, weak Alpha, Sparse Rips, Cubical) × 7 vectorizers (Persistence Image, Persistence Landscape, Betti Curve, Persistence Statistics, Silhouette, Amplitude, Persistence Entropy) × 4 classifiers (SVM-RBF, SVM-linear, Random Forest, Logistic Regression) — 672 possible; we exclude the 56 configurations pairing `weak_alpha` or `sparse_rips` with MNIST — weak Alpha because Delaunay triangulation of a 28-point cloud in $\mathbb{R}^{28}$ is infeasible, and Sparse Rips because its giotto-tda implementation fails on the same input Vietoris–Rips accepts — and the remaining 616 completed. All filtrations computed H_0 and H_1 homology. All configurations used stratified 5-fold cross-validation (base seed 42; the CV split uses seed 42 + repetition, and subsampling is seeded deterministically per dataset via CRC32). Point clouds were generated at 100 points per cloud (200 samples per noise level); uniform random subsampling is applied by the runner only if a source cloud exceeds the target size.

Datasets. Six dataset instances span three modalities. Sphere/torus point clouds appear at four noise levels ($\sigma \in \{0.00, 0.05, 0.15, 0.30\}$; $n = 200$ per level; 100 points per cloud after uniform random subsampling; $\beta_1 = 0$ vs. 2). ECG200 heartbeat recordings (normal vs. myocardial infarction; $n = 200$; 96 timesteps) are Takens-embedded to 94×3 arrays with $d = 3, \tau = 1$ and interpreted as point clouds by Vietoris-Rips, weak Alpha, and Sparse Rips, and as a 94×3 pixel grid by CubicalPersistence — the latter is *not* a point-cloud filtration; see the disclosure below. ECG200 is a UCR archive benchmark.

Under the 25-repetition repeated-CV protocol, raw-signal baselines (logistic regression 85.28%, random forest 86.30%) beat our best TDA configuration (83.0%, a single-split point estimate); we therefore do *not* claim that TDA features are competitive with classical methods on this dataset — the sweep is an internal TDA-pipeline com-

parison (measured baselines are reported in §4.1). Note also that ECG200 is energy-normalised: the sum of squares of every record is constant (95.00 across all 200 signals), so the trivial-separator concern that would attach to an un-normalised signal does not apply to the executed data.

The repeated-CV-stable best configuration and the effect of excluding the cubical filtration are quantified in §4.1; the sweep is an internal TDA-pipeline comparison, not a competition with classical baselines (§4.1.1).

MNIST binary. To test image-specific filtrations, a binary subset of MNIST (digits 0 vs. 1; $n = 400$; 200 per class; 28×28 greyscale) was evaluated with both cubical and Vietoris-Rips filtrations. Cubical filtration achieved 98.0% accuracy (with Betti Curves + Logistic Regression), versus 96.25% for the best Vietoris–Rips configuration: a ∼1.75pp best-of-family gap. However, the §4.1 marginal analysis shows the vectorizer moves accuracy more than the filtration on this dataset (3.22pp versus 1.65pp): on this binary subset the filtration still matters, but it is not the stage that limits accuracy. This is consistent with Conti et al.'s demonstration that a badly-matched filtration can be catastrophic at scale (18% to 94% on 10-class MNIST under grid search) — but the effect is much smaller on our binary subset, where even the worst filtration remains informative.

Table 4.1: Dataset instances used in the benchmark sweep. Point clouds contain 100 points; ECG200 has 96 timesteps. The torus has major radius $R = 2$ and minor radius $r = 1$ (so point norms range $[1, 3]$), while the sphere has unit radius (norms $\equiv 1$): the classes differ in scale as well as topology.

Dataset	Modality	n (samples)	Dims	Classes	Maj.%	Source
Sphere/Torus (σ=0.00)	point cloud	200	100 pts	2	50.5%	synthetic
Sphere/Torus (σ=0.05)	point cloud	200	100 pts	2	50.5%	synthetic
Sphere/Torus (σ=0.15)	point cloud	200	100 pts	2	50.5%	synthetic
Sphere/Torus (σ=0.30)	point cloud	200	100 pts	2	50.5%	synthetic
ECG200	time series	200	96 ts	2	66.5%	UCR archive [13]
MNIST 0/1	image	400	28×28	2	50.0%	MNIST subset

4.1 Stage Variance Analysis

Which pipeline stage contributes the largest variance? For each stage, we marginalise over the other two and compute the range of marginal accuracies from the repeated-

CV means (25 repetitions, seeds 43–67, for ECG200; 5 repetitions, seeds 43–47, for MNIST), with 95% CIs over the per-split ranges, and decompose the variance with an η^2 analysis (fold-level F-tests) to test whether the range exceeds sampling noise.

Table 4.2: Marginal accuracy by pipeline stage on ECG200 (repeated CV, 25 repetitions).

Stage	Top Performer	Marg. Acc.	Range (95% CI)	p-value
Filtration	Cubical	73.15%	0.69pp [0.57, 0.81]	0.93
Vectorizer	Landscapes	75.32%	6.39pp [6.13, 6.65]	9.7×10^{-5}
Classifier	Random Forest	75.29%	3.50pp [3.28, 3.71]	0.004

Ranges, CIs, and marginal accuracies are repeated-CV values (25 repetitions of stratified 5-fold CV, seeds 43–67; Sparse Rips dropped for runtime, so the config space is 84 configurations): the range is the mean over repetitions of the per-split marginal range, with a 95% t-interval across the 25 per-split ranges (repeated-measures CI; Nadeau–Bengio corrected resampled CIs are wider: vectorizer [5.69, 7.10], classifier [2.92, 4.07], filtration [0.37, 1.01], all excluding zero; the correction is that of Nadeau and Bengio [24]). p-values are F-test p-values from the per-configuration three-way ANOVA (Table 4.3); the fold-level panel of that table is descriptive only, as folds within a configuration share training data, so its p-values are anti-conservative. No multiplicity correction is applied across the three stage comparisons.

The full menu favours vectorization: its range (6.39pp across 7 vectorizers) is roughly nine times larger than filtration (0.69pp across 3 filtrations in the 25-repetition sweep; 0.70pp across 4 including Sparse Rips at r=5), and the ordering vectorizer $>$ classifier $>$ filtration was stable across all 25 CV repetitions. These intervals quantify split-resampling noise on this sample; they are not estimates of sampling variance across the underlying population. This ordering is significant under non-parametric inference on the 25 repetition-level stage rankings. The Friedman statistic is $Q = \frac{12n}{k(k+1)} \sum_j (\bar{R}_j - \frac{k+1}{2})^2 = 50$ for k=3 stages and n=25 repetitions (asymptotic χ^2-tail $p \approx 1.4 \times 10^{-11}$). The exact permutation value ($p \approx 2 \times 10^{-19}$) and the pairwise sign test (25/25 repetitions; $p = (1/2)^{25}$) are *descriptive*: the 25 repetitions are resamples of one 200-sample dataset, not independent replicates, so these p-values quantify ranking consistency across resamples, not significance against a dataset population. The variance decomposition of Table 4.3 is the primary evidence for this ordering: on ECG200 the vectorizer explains 21.7% of the per-config variance (fold-level 10.6%), the classifier 9.8% (4.8%), and the filtration 0.3% (0.1%). The top-performing vectorizer, Persistence Landscapes, achieves 75.32% marginal accuracy versus 68.95% for Persistence Entropy (the weakest). Cubical filtration leads at 73.15%, with Vietoris-Rips (72.89%) and Weak Alpha (72.74%) within 0.41pp.

Equal-footing re-analysis. The ranges above compare stage menus of different sizes (7 vectorizers versus 3 filtrations on ECG200) and include "degenerate" scalar vectorizers — Persistence Entropy and Amplitude, each of which maps a diagram to

a single number — that are structurally handicapped. To test whether the 6.39pp (ECG200) and 24.89pp (ECG5000) vectorizer ranges are an artefact of that handicap, we re-measure the stage effects on an equal footing: under the design-penalised ω^2, and by re-computing the marginal ranges excluding the degenerate scalar vectorizers and matching the number of levels.

By ω^2 the vectorizer remains the largest single stage main effect on both real datasets (ECG200 0.165 versus classifier 0.077 and filtration −0.017; MNIST 0.214 versus 0.143 and 0.032); the level counts do not manufacture the ordering. The marginal-range *size*, however, is: excluding Amplitude and Persistence Entropy cuts the ECG200 vectorizer range from 6.39pp to 3.10pp — and the same exclusion applied symmetrically to all three stages puts the *classifier* first (vectorizer 3.10pp, classifier 3.87pp, filtration 1.94pp; the classifier is the largest stage in 20 of the 25 repetitions; `repeated_cv_r25.db`, per-repetition mean method). The full-menu ordering rests on floor effects in two menus at once (scalar vectorizers and the SVM-RBF collapse). On the ECG5000 probe the 24.89pp range collapses to 0.24pp (Persistence Entropy at 36.9% versus Betti Curve 61.5% and Silhouette 61.8%, which are within 0.24pp of each other). Matching the number of levels — best-3 vectorizers versus 3 filtrations on ECG200 — narrows the vectorizer edge to 1.21pp versus 0.69pp; on MNIST the best-2 vectorizer range (0.44pp) falls below the 2-filtration range (1.55pp), and on ECG5000 the best-2 vectorizer range (0.24pp) likewise falls below the filtration range (3.60pp).

The isolated contribution — swapping one stage while holding the other two at their marginal-best — is still largest for the vectorizer on ECG200 (10.22pp, r=25 non-degenerate slice; `repeated_cv_r25.db`) and MNIST (2.1pp, r=5 non-degenerate slice). This isolated-contribution metric holds the other two stages at their best levels, which maximally advantages the varied stage; it is an upper bound on the stage's achievable range rather than a typical operating difference. The defensible conclusion is that vectorization is the largest single stage effect under the full menus (and by ω^2 after adjusting for interactions), but its measured *margin* is sensitive to the menu and to the number of levels compared: the 6.39pp and 24.89pp ranges are inflated by degenerate scalar vectorizers; under symmetric scalar exclusion the classifier leads (3.87pp vs. 3.10pp vectorizer), and at matched level counts the vectorizer-vs-filtration gap narrows (ECG200) or reverses (MNIST, ECG5000).

Simple versus complex. Persistence Statistics matches kernel-based methods on ECG200: its r=5 marginal accuracy (74.8%) is within 0.3pp of Persistence Landscapes (75.1%), and it is best or tied in 4 of the 16 r=5 filtration–classifier cells (all under SVM-linear; pooled over the five repetitions), ranking above the median vectorizer in most cells. Statistics likewise leads Persistence Images (72.9%). On MNIST binary, Persistence Statistics reaches 95.9% marginal accuracy, within 2.1pp of the best configuration overall (Cubical + Betti Curve + Logistic Regression at 98.0%, Table 4.7).

The evidence base remains small ($n = 3$ real datasets: two time series and one image, with the repeated-CV verification covering ECG200 only), and the corrected intervals above should not be read as licence for fine-grained ranking. Per-configuration accuracy varies by 1.09pp on average across the five splits (maximum 3.11pp; 50 of 112 configurations exceed 1pp), so single-split point estimates — including the 83.0% best configuration of Table 4.7, whose repeated-CV mean is 79.6% — carry ±1–3pp of split-to-split noise.

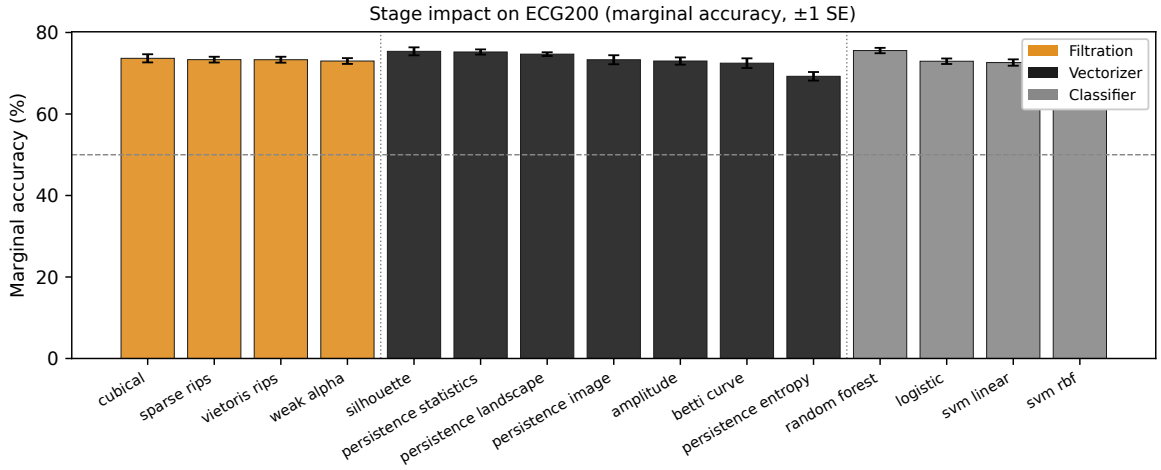


Figure 4.1: Stage impact on ECG200. Error bars show ±1 SE.

All 112 configurations on the synthetic sphere/torus data achieve $\geq 99.5\%$ accuracy at $\sigma = 0.00$ (mean 99.99%). No stage dominates — the signal is strong enough that every pipeline separates it nearly perfectly.

MNIST binary marginal analysis. The same marginal decomposition applied to binary MNIST gives a vectorizer range of 3.22pp versus a filtration range of 1.65pp: vectorization is the larger stage even on image data. The ordering persists under repeated 5-fold CV (5 repetitions, seeds 43–47; `data/tda/mnist_repeated_cv.db`): it holds in all five repetitions (vectorizer 2.91–3.22pp vs. filtration 1.49–1.65pp, pooled 3.03pp vs. 1.55pp), and the best configuration (cubical + Betti Curve + Logistic, 98.0%) is stable across repetitions (97.5–98.0%). Filtration effects are real but small — cubical beats Vietoris-Rips by ∼1.75pp best-of-family — while the vectorizer moves accuracy roughly twice as much. Table 4.3 reports the complementary variance decomposition (η^2: the proportion of accuracy variance explained by each stage): on ECG200 the vectorizer explains 0.217 of the variance versus 0.098 for the classifier and 0.003 for the filtration; on MNIST the vectorizer leads (0.302) over the filtration (0.166). The MNIST filtration F exceeds the vectorizer F (15.83 vs. 4.80) despite lower η^2 (0.166 vs. 0.302), because F depends on the degrees of freedom (filtration df=1, vectorizer df=6); η^2 is the ordering statistic. Both analyses agree: vectorization is the dominant stage on both real datasets, and the filtration contribution is dataset-specific and

modest. The ω^2 population effect size (corrected for the number of levels; recomputed from the 25-repetition ECG200 sweep, 84 configurations) confirms the ordering under the design penalty: on ECG200 the vectorizer's $\omega^2 = 0.165$ (bootstrap 95% CI [0.063, 0.423]) versus classifier 0.077 ([0.003, 0.246]) and filtration -0.017 ([-0.017, 0.093]). On MNIST, vectorizer 0.214 ([0.103, 0.540]) versus filtration 0.143 ([0.040, 0.312]) and classifier 0.032 ([-0.015, 0.257]). The vectorizer ω^2 CI excludes zero on both real datasets, while the filtration CI on ECG200 straddles zero.

Full 10-class MNIST. The binary-MNIST ordering is not scale-invariant. Rerunning the same protocol on the full 10-class MNIST set (1000 samples, 100 per class; cubical and Vietoris-Rips filtrations, 4 vectorizers, Random Forest and SVM-RBF, 5-fold CV reps 43–47; `data/tda/mnist10_sweep.db`) flips the stage ordering: the filtration marginal range (4.53pp: cubical 29.1% versus Vietoris-Rips 33.7%) now exceeds the vectorizer range (3.44pp: Betti Curve 33.4% to Landscapes 30.0%) and the classifier range (2.03pp). Within a single filtration the configuration span (vectorizer $\times$ classifier) is larger — cubical 10.40pp, Vietoris-Rips 9.10pp (both pooled over repetitions) — so the stage-dominance question is filtration-dependent (an interaction, §4.1). Absolute accuracy is much lower than binary MNIST (best 37.8%; chance 10%), because ten-way separation is intrinsically harder. This is consistent with Conti et al.'s emphasis on filtration for multiclass images, but the swing we observe (4.53pp marginal; 9–10pp within a filtration) is far smaller than their reported 18%–94%, which used grid search, 10-split CV and a broader filtration family. The binary-MNIST ordering is a binary-scale phenomenon: at 10-class scale, filtration becomes the larger marginal stage.

Table 4.3: η^2 variance decomposition of classification accuracy by pipeline stage. Per-config panel: classic $\eta^2 = \mathrm{SS}_{\mathrm{stage}}/\mathrm{SS}_{\mathrm{total}}$ over per-configuration means (one observation per configuration). Fold-level panel: main-effects-only three-way ANOVA over the fold accuracies of the single-split sweep, with classic η^2 and F-tests.

Dataset	Stage	Per-config means			Fold-level		
		η^2	F	p	η^2	F	p
ECG200	Filtration	0.003	0.15	0.93	0.001	0.32	0.81
ECG200	Vectorizer	0.217	5.26	9.7×10^{-5}	0.106	11.41	4.9×10^{-12}
ECG200	Classifier	0.098	4.75	0.004	0.048	10.31	1.3×10^{-6}
MNIST 0/1	Filtration	0.166	15.83	2.5×10^{-4}	0.061	20.47	9.1×10^{-6}
MNIST 0/1	Vectorizer	0.302	4.80	7.3×10^{-4}	0.111	6.21	4.1×10^{-6}
MNIST 0/1	Classifier	0.061	1.95	0.135	0.023	2.52	0.058
Sphere/Torus (σ=0.00)	Filtration	0.009	0.41	0.748	0.002	0.34	0.793
Sphere/Torus (σ=0.00)	Vectorizer	0.165	3.67	0.002	0.032	3.10	0.005
Sphere/Torus (σ=0.00)	Classifier	0.083	3.67	0.015	0.016	3.10	0.026

Per-config scope: $N = 112$ (ECG200, sphere/torus) or 56 (MNIST) configurations; fold-level scope: $N = 560$ or 280 fold accuracies. Degrees of freedom: filtration 3 (MNIST 1), vectorizer 6, classifier 3. The repeated-CV analysis of Table 4.2 corroborates the ECG200 ordering. *Multiplicity:* the 18 F-tests in this table (3 datasets $\times$ 3 stages $\times$ 2 scopes) were recomputed from `data/tda/expanded_results.db` (agreement with the published `eta_squared_results.csv` to machine precision) and adjusted for multiplicity. Holm (FWER) leaves 10/18 significant at α=0.05; Benjamini–Hochberg (FDR) leaves 12/18. The headline results — ECG200 fold-level vectorizer ($p = 4.9 \times 10^{-12}$, Holm 8.7×10^{-11}), MNIST vectorizer and filtration — survive both corrections; the marginal filtration effects that fail are the already-weak sphere/torus and ECG200 filtration rows. The later expansion sweeps each carry their own comparisons without global multiplicity correction; most are estimation rather than hypothesis testing, which we judge defensible.

Two-way interactions qualify the stage decomposition. The η^2/F analysis of Table 4.3 is main-effects-only: each stage's contribution is assessed in isolation. Because the factorial design varies all three stages jointly, we also fitted a three-way ANOVA with all two-way interaction terms to the fold-level accuracies of the repeated-CV sweeps (`repeated_cv_r25.db` for ECG200, N=10500 fold accuracies; `mnist_repeated_cv.db` for MNIST, N=1400). The ECG200 two-way interactions are not negligible: the three interaction terms jointly carry $\omega^2 = 0.187$, which is *147% of the main-effect total* (0.127). The filtration–vectorizer interaction ($\omega^2 = 0.092$) is the *largest single effect in the model*, exceeding the vectorizer main effect (0.086); the vectorizer–classifier interaction (0.089) rivals it. On MNIST the interactions carry 50% of the main-effect total (0.103 versus 0.206), led by vectorizer–classifier (0.054) and filtration–vectorizer (0.047). Fold-level p-values in this analysis are anti-conservative (folds within a configuration share training data), so the effect sizes — the ω^2 values — are the robust quantities and the p-values are indicative. The practical implication

is that the stages are *not cleanly separable*: the effect of changing one stage depends on the levels at which the other two are held. Vectorization remains the largest single *main* effect on both datasets, so the direction of the headline finding is unchanged, but the magnitude of the interaction terms shows that "which stage matters most" is partly conditional on the other two stages, not a stage-intrinsic property. We therefore read the stage-dominance claim as *dominance in main effect*, and treat the interaction structure as an explicit boundary condition on that claim.

Repeated cross-validation. Because single-split point estimates are noisy (per-configuration SD across repetitions averages 1.27pp, max 3.33pp at 25 repetitions), we re-ran the ECG200 analysis with repeated CV (25 repetitions of the same stratified 5-fold protocol, seeds 43–67). Table 4.2 and Figure 4.1 report the repeated-CV ranges and CIs; the ordering vectorizer > classifier > filtration was stable in all 25 repetitions. Increasing from r=5 to r=25 narrows the repeated-measures CI widths by ∼3× (filtration 0.69pp → 0.24pp, classifier 1.20pp → 0.43pp) at no change to the ordering. By 25-repetition mean, cubical + Persistence Image + Random Forest is the best ECG200 configuration (82.82%, r=25; the same configuration averages 83.2% at r=5); the single-split point estimate for cubical + Silhouette + Random Forest is 83.0% (Table 4.7). Differences below ∼1pp between individual configurations should not be over-interpreted.

Generality probe: ECG5000. The ECG200 result could be instance-specific, so the stage analysis was repeated on a second UCR recording, ECG5000 (UCR archive; 5 classes; 140 timesteps; single patient recording, BIDMC chf07). The same 5-fold protocol was applied to a stratified subsample of 714 samples ($\leq$ 200 per class) drawn from the full 5000-sample recording (class counts 2919, 1767, 96, 194, 24; majority class 58.38% in the full recording; the executed 714-sample subsample is balanced to $\leq$ 200/class, so its majority share is 28.0%). The pipeline subset comprised 2 filtrations, 3 vectorizers, and 2 classifiers: Persistence Entropy (the weakest ECG200 vectorizer) plus Silhouette and Betti Curve (3rd and 5th of 7 at r=25); the ECG200 winner, Persistence Landscapes, is absent, so the probe bounds the floor of the menu, not its ceiling. Two of the twelve configurations fail on a technicality (silhouette-on-weak-alpha produces NaN features) and are excluded. Both failures are weak-alpha cells, so the weak-alpha marginal is computed over 4 configurations while the Vietoris-Rips marginal uses 6 including silhouette — the filtration range is computed on non-matching vectorizer menus. Because the classes are severely imbalanced, we report balanced accuracy and macro-F_1 alongside plain accuracy. The result replicates the hierarchy: vectorizer marginal range 24.89pp versus filtration 3.60pp (plain accuracy, point estimate, single split); balanced accuracy per vectorizer is 0.46–0.47 (Betti Curve, Silhouette) versus 0.28 (Persistence Entropy) and macro-F_1 0.43–0.45 (per-vectorizer marginals; the best cells reach 0.50) versus 0.12–0.38 for Persistence Entropy, so vectorization leads under

class-balanced metrics too. The vectorizer-led ordering on time series does not depend on the specific ECG200 instance; a full multiclass generalisation claim would require multiple datasets (the probe is single-patient).

Beyond accuracy. Plain accuracy on ECG5000 is dominated by the majority class (58.38% in the full recording; 28.0% in the balanced subsample). Re-running the grid and capturing per-fold probabilities (AUROC one-vs-rest macro, per-class precision/recall/F_1, multiclass Brier; `data/tda/beyond_accuracy_ecg5000.db`) shows three things. First, vectorization dominance holds on the rank scale: the vectorizer AUROC marginal range is 19.38pp (Betti Curve 77.6%, Silhouette 77.5%, Persistence Entropy 58.2%) versus 11.90pp for the classifier and 3.31pp for filtration — the vectorizer is *more* dominant under a class-imbalance-robust metric than under accuracy, because Persistence Entropy (a degenerate scalar) is worst on AUROC too. Second, the classifier range is driven by SVM-RBF's near-chance AUROC (0.47) on the entropy feature, i.e. the RBF classifier collapsed to the majority class there. Third, per-class recall exposes a severe imbalance failure that accuracy hides: the mean of per-(configuration, fold) per-class recalls shows that the minor classes are largely unlearned (class 2, 96 samples, 16.3% recall; class 3, 194 samples, 31.8%; class 4, 24 samples, 1.7%, with Betti Curve and Silhouette at 0%), while the majority classes 0 and 1 are recovered at 73.9% and 71.8%. So the reported accuracy (65.8% best configuration) is carried almost entirely by the two largest classes; on extreme imbalance the TDA pipeline, like most discriminative classifiers, defaults to the majority. Calibration is likewise classifier-dependent: the multiclass Brier for the best configs (Random Forest) is 0.46–0.49 versus 0.55–0.75 for SVM-RBF (Platt-scaled), so Random Forest is both better ranked and better calibrated here.

Cubical robustness check. CubicalPersistence on ECG200 operates on the 94×3 Takens-embedded array as a pixel grid (intensities are treated as coordinates), not on a point-cloud filtration. Excluding cubical from the ECG200 analysis leaves the hierarchy intact: the best non-cubical configuration (Vietoris-Rips + Persistence Entropy + Random Forest) reaches 79.5% (single-split), the vectorizer range widens slightly to 6.58pp, and the filtration range drops to 0.34pp.

Cubical grid-structure ablation. Because cubical persistence reads the 94×3 array as a greyscale pixel grid, we ablated the grid layout directly: independently permuting the 94 rows (time-window order) or the 3 columns (Takens coordinates) of each sample, labels intact, and re-running all 28 cubical configurations (CV seeds as the sweep). Row shuffling drops the mean accuracy from 73.66% to 71.96% and collapses the top grid-sensitive vectorizers to the majority class: Betti Curve + SVM-RBF falls from 80.5% to 66.5% and the Persistence Image family from 76.5–82.0% to 64.0–76.5%. Column shuffling is milder (mean 72.45%; the Takens-coordinate order carries less information than the time-window order), and Persistence Landscape is the

vectorizer whose accuracy rises most under row shuffling (72.5–80.0% unshuffled, mean 75.5, vs. 79.0–83.5% shuffled, mean 82.4; +6.9pp; Silhouette also rises, +2.1pp, while every other vectorizer falls), so it is grid-antagonistic rather than grid-driven — but it is not permutation-invariant. The cubical result is partly grid-structure-driven: the time-window layout of the embedded signal contributes real signal for the top vectorizers, so the cubical-on-grid accuracy should not be read as a pure point-cloud-topology result (consistent with the disclosure above).

4.1.1 Classical Baselines and the Role of Raw Features

To calibrate the absolute accuracy of the TDA pipelines, we ran non-topological baselines under the same stratified 5-fold CV protocol as the TDA repeated-CV analysis (25 repetitions, CV seeds 43–67, so the baseline CIs are directly comparable to the TDA numbers of §4.1):

Table 4.4: Non-topological baselines (25-repetition repeated CV, mean ± 95% CI over repetitions). Majority-class rows give the trivial-separator floor. TDA best rows are single-split point estimates.

Dataset	Approach	Classifier	Mean acc.	95% CI
MNIST 0/1	majority class	—	50.0%	—
MNIST 0/1	raw pixels (784)	Logistic	99.65%	[99.60, 99.70]
MNIST 0/1	raw pixels (784)	Random Forest	99.29%	[99.19, 99.39]
MNIST 0/1	TDA best	cubical + Betti + Logistic	98.0%	—
ECG200	majority class	—	66.5%	—
ECG200	raw signal (96)	Logistic	85.28%	[84.92, 85.64]
ECG200	raw signal (96)	Random Forest	86.30%	[85.68, 86.92]
ECG200	TDA best	cubical + Silhouette + RF	83.0%	—

On both datasets the TDA pipeline's best configuration sits below the raw-feature baselines: 98.0% versus 99.65% (logistic regression) on MNIST binary, and 83.0% (single-split point estimate) versus 85.28% (logistic) / 86.30% (random forest) on ECG200 under the identical 25-repetition protocol. All TDA and baseline results comfortably exceed the majority-class floor (50.0% on MNIST binary, 66.5% on ECG200). The claim that "TDA features can be competitive with classical methods" is scoped to the internal TDA comparison: we make no claim of beating raw features on these datasets. The noise experiments (§4.2) also show that a single per-cloud mean-norm feature (logistic) separates the synthetic sphere/torus classes at 100% at every noise level (a fixed-threshold hand rule on the norm fails at σ=0.30: 50.5%), which quantifies the scale confound and motivates the matched-genus experiment reported there.

Repeated-CV harmonisation. ECG5000, the matched-genus control, and the panel were evaluated under the same 25-repetition protocol (5-fold, seeds 43–67; the fixed panel grid: Vietoris–Rips and DTM-weighted Rips × three vectorizers × two classifiers), adding 1,500 runs (`data/tda/r25_ecg5000.db`, `r25_genus.db`, `r25_panel.db`; all finished).

On ECG5000 the repeated-CV marginal ranges put the vectorizer first at 4.30 pp (95% CI [4.02, 4.57]), ahead of the classifier (2.96 pp [2.81, 3.11]) and the filtration stage (0.96 pp [0.80, 1.13]); the ordering holds in 24 of 25 repetitions, with all three CIs excluding zero. Under the fixed panel grid and repeated CV the vectorizer dominates with disjoint confidence intervals; the magnitude depends on the menu, the ordering does not. On the matched-genus control the vectorizer leads, at 3.89 pp ([3.49, 4.30]), while the filtration (0.71 pp) and classifier (0.41 pp) ranges stay small, and the 25-repetition mean accuracies are 98.35% at σ=0 and 90.98% at σ=0.30 for the VR/DTM-weighted grid (the single-split matched-genus analysis of §4.2 used a different menu, VR/weak-alpha; 95.83% at σ=0.30).

On the two-dataset panel subset, the pooled ranges are dominated by the cross-dataset modality composition (filtration 43.30 pp, dataset 41.22 pp — the time-series arm runs VR/DTM while the image arm runs cubical/VR), so the interpretable quantity there is the within-dataset ordering, which reproduces the vectorizer-leads pattern of §4.1.

On **image data**, the raw input is a 2D grid of pixel intensities. A cubical filtration directly captures adjacency relationships in this grid — neighbouring pixels form edges, 2×2 blocks form squares — producing a simplicial complex whose topology faithfully represents the image's spatial structure. A point-cloud filtration like Vietoris-Rips, applied to the same data (which the pipeline feeds to the filtration as its native array — for MNIST, 28 rows of 28 intensities, i.e. a 28-point cloud in $\mathbb{R}^{28}$), does not see the 2D grid adjacency at all: the filtration's notion of "nearby" is Euclidean distance between rows, so the pixel-grid topology is absent from the outset.

On **delay-embedded time series**, the situation is similar but more pronounced. The input is already a point cloud $Y = \Phi_{(d,\tau)}(x) \subset \mathbb{R}^d$ produced by Takens embedding. Vietoris-Rips and weak Alpha complexes operate on the same metric space (Y, d_E); their filtrations are not identical — VR (Vietoris–Rips) and the full Alpha complex are multiplicatively interleaved, which bounds their bottleneck distance rather than equating their diagrams — and the empirical filtration range on ECG200 is just 0.69pp (25-repetition repeated-CV mean). The embedding captures the dynamical structure; the filtration reads it out.

4.2 Robustness of the Ordering

Does topological signal survive additive spatial noise? Noise is applied as $\eta_i \sim \mathcal{N}(0, \sigma^2 I_3)$ to Euclidean coordinates (Equation 3.3). Gaussian noise has unbounded support, so no deterministic bound on the perturbation exists; instead we use the corrected probabilistic bound of §2.4 (Equation 2.9). That bound is a typical-scale estimate whose threshold the perturbation maximum exceeds with probability $\approx 93\%$ at d=3 (§2.4), so the empirical check is the measured d_B itself, which stays far below the bound (max 0.300 at σ=0.15, max 0.434 at σ=0.30; the $2\sigma\sqrt{2 \ln n}$ values are 0.91 and 1.82 for n=100).

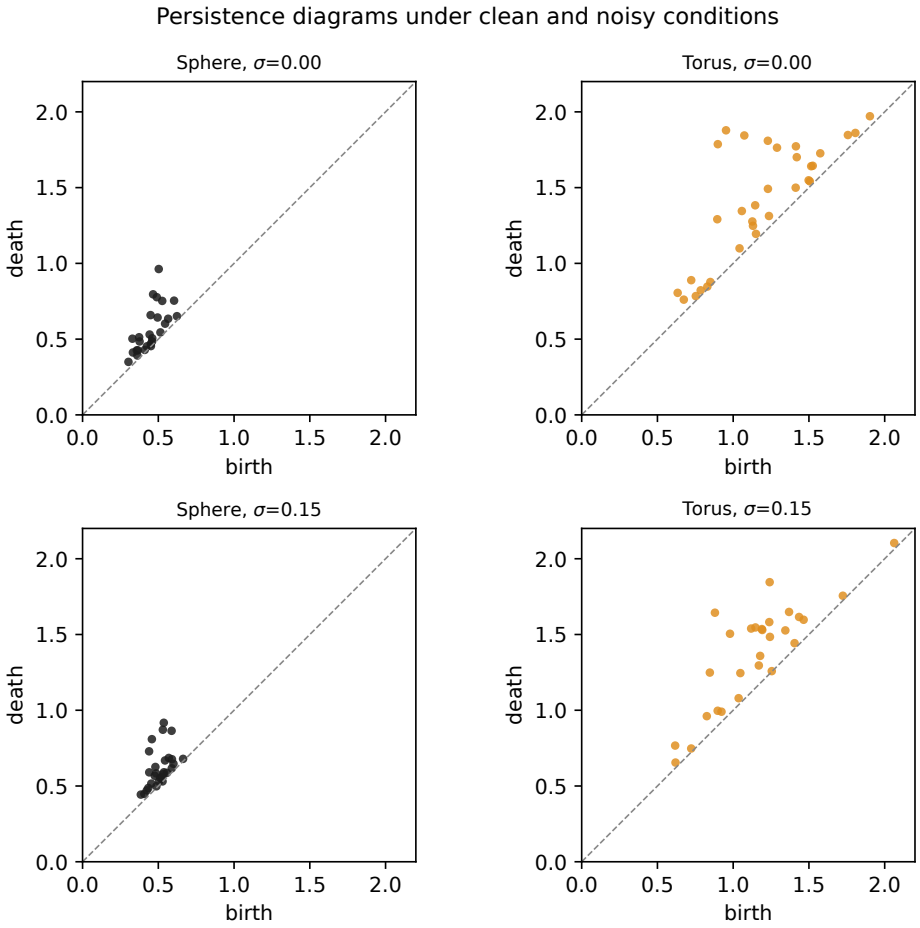


Figure 4.2: Persistence diagrams under clean and noisy conditions.

At $\sigma = 0.00$, 0.05, and 0.15, every configuration achieves $\geq 99.5\%$ accuracy (mean 99.99%, 100.00%, and 99.97% respectively). The stability bound at $\sigma = 0.15$ is $d_B \lesssim 0.91$ (Equation 2.9). Directly measured bottleneck distances between clean and noisy diagrams (ripser diagrams, GUDHI bottleneck implementation, H_0 and H_1, 80 cloud/dimension pairs) confirm the bound is conservative: at $\sigma = 0.15$ the mean measured d_B is 0.144 and the maximum 0.300, both well below the bound.

At $\sigma = 0.30$ (bound $\lesssim 1.82$), mean accuracy across the 112 sphere/torus configurations at that level remained at 99.85% (minimum 98.5%). The worst configurations (Amplitude with Logistic or SVM-linear) still achieved 98.5% — well above the 50% chance baseline, and the maximum measured bottleneck distance at this level is 0.434, less than a quarter of the corrected bound. The torus H_1 diagrams move on average by $d_B = 0.283$ at this level — below the measured H_1 persistence range [0.556, 1.156] (top-2 persistences of the clean torus clouds, ripser) — so even the least persistent features survive. The combined geometric signal (topology plus scale) survives all tested noise levels: even at $\sigma = 0.30$, the $\beta_1 = 0$ vs. $\beta_1 = 2$ distinction is separated by linear kernels at $\geq 98.5\%$ in every vectorized feature space tested.

Isolating the topological signal. The sphere/torus classes differ in scale as well as topology (torus point norms in $[1, 3]$ versus sphere $\equiv 1$): a single per-cloud mean-norm feature (logistic) separates them at 100% at every noise level (a fixed-threshold hand rule on the norm fails at σ=0.30: 50.5%), so the survival results above describe the *combined* geometric signal. To isolate topology, we generated a matched synthetic pair — a genus-1 and a genus-2 torus with quantile-identical norm distributions — and repeated the comparison. Norm-based features now collapse to near chance (48–58%; the clean-level 58% is within $\sim$2 SE of chance at $n = 200$), while the TDA pipeline separates the pair at 99.75% clean and 95.83% at $\sigma = 0.30$ (minimum 91% over configurations). Caveat: the matched pair is still linearly separable in raw coordinates (logistic regression on raw coordinates achieves 100% at both $\sigma = 0.00$ and $\sigma = 0.30$, `baseline_experiments.db`); the experiment isolates the norm/scale confound specifically, not linear separability in general. The TDA result (95.83% at $\sigma = 0.30$) shows the topological signal survives the norm/scale confound control and remains robust to noise at the levels tested. (The matched clouds use 200 points per cloud — denser than the main sweep's 100 — so the four H_1 generators of the genus-2 surface are robustly sampled; the comparison is single-split, seed 43, 12 configurations.)

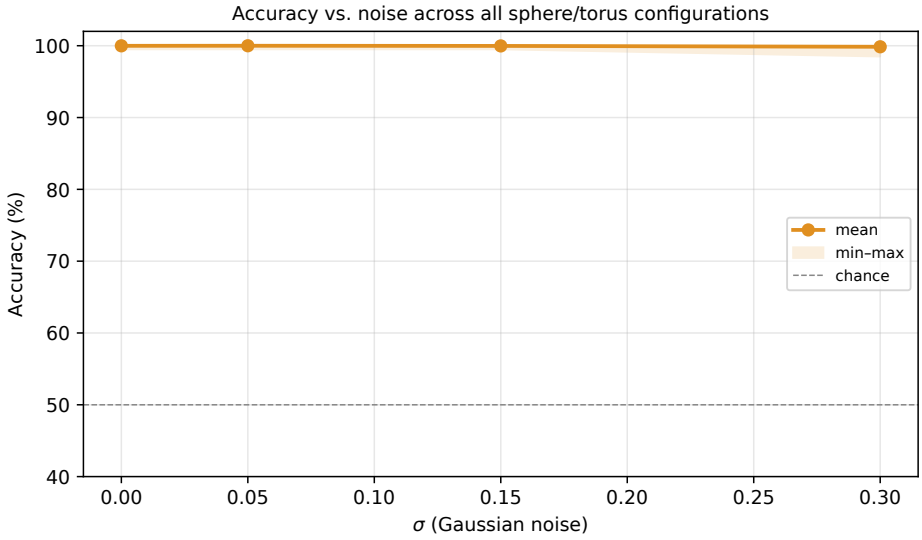


Figure 4.3: Accuracy vs. noise (per-configuration mean across the 112 sphere/torus configurations at each level; band = min–max spread). All pipelines maintain $\geq 98.5\%$ accuracy through $\sigma = 0.30$; measured bottleneck distances (max 0.434 at $\sigma = 0.30$ versus the corrected bound $2\sigma\sqrt{2 \ln n} \approx 1.82$) confirm the stability bound is conservative.

Diverse-filtration check. The filtration range above is over a menu in which two of three filtrations are Rips-type (Vietoris-Rips, weak Alpha; Sparse Rips at r=5), all approximating the same Rips geometry, so the empirical filtration range is sensitive to the filtration menu. Adding a genuinely different filtration — the DTM- weighted Rips (giotto-tda `WeightedRipsPersistence`, vertex and edge weights from a distance-to-measure estimate; Anai et al. [3]), robust to outliers — raises the ECG200 filtration marginal range to 2.81pp and makes DTM the best filtration (76.2% versus 73.4% for Vietoris-Rips), winning 7 of 8 vectorizer–classifier pairs (best configuration DTM + Landscapes + Random Forest, 81.0%; `data/tda/filtration_diversity_sweep.db`, 48 runs (16 ECG200 configurations), single split). Vectorization remains the largest stage (5.62pp), so the headline ordering survives; but the empirical filtration range is not a universal constant — it is sensitive to the filtration menu, and a genuinely non-Rips filtration both matters (2.81pp) and performs better than the Rips-type baseline. On the clean synthetic sphere/torus pair all 16 diverse-pool configurations reach 100%, regardless of filtration, so no stage matters there (at σ=0 the vectorizer effect is statistically non-zero but < 0.1pp on a near-ceiling task; Table 4.3).

Panel with a diverse filtration set. We re-ran the nine-dataset panel with more than one *working* filtration per modality so that the filtration stage and the vectorization stage could be compared on equal footing (each stage now has at least two alternatives). For the seven time-series datasets the two filtrations were Vietoris–Rips and the DTM-weighted Rips filtration (`WeightedRipsPersistence` with `weights='DTM'`, a genuinely non-Rips construction); weak alpha was excluded from the time-series

arm because it does not complete on the quantized series (Wafer, ElectricDevices, ECG5000), which is itself a portability result.

For the two image datasets the filtrations were cubical and Vietoris–Rips. The four vectorizers (persistence image, persistence landscape, Betti curve, silhouette) and two classifiers (RBF-SVM, random forest) were unchanged, and evaluation used the same 5-fold stratified CV with seed 42 (rep 1). Across the nine datasets of the stage-capable panel (`panel_stagecapable.db`; two working filtrations per modality), with each range computed as the spread of stage-level means of per-config fold-mean accuracies, the vectorization range exceeds the filtration range on 7 of the 9 datasets; only HandOutlines (1.90 vs. 4.22 pp) and mnist10 (2.80 vs. 4.41 pp) favour filtration.

Across datasets the filtration range spans a minimum of 0.31 pp, a median of 1.37 pp, and a maximum of 4.41 pp (mean 2.17 pp), while the vectorization range spans a minimum of 0.62 pp, a median of 3.32 pp, and a maximum of 15.00 pp (mean 5.11 pp). Vectorization still leads on average, roughly 2.4× the filtration range; however, the comparison is now built on a genuinely diverse non-Rips filtration rather than Vietoris–Rips alone, and the filtration range is no longer negligible (the previous single-filtration panel fixed that stage, making its contribution invisible).

Restricting the vectorizer to its two best performers on the panel (Betti curve, persistence image) for a level-matched two-versus-two comparison, the mean vectorizer range falls to 2.02pp (median 1.90pp; min 0.50pp, max 5.78pp) — of the same magnitude as the 2.17pp filtration range (ratio 0.93×) — and the vectorizer exceeds the filtration range on 5 of the 9 datasets. The 2.4× gap is a level-count artefact: at matched numbers of alternatives per stage the panel shows no dominant stage on average, consistent with the population-level mixed model below and with the conditionality thesis of Chapter 6.

The vectorization lead is concentrated in the range's upper tail: removing the single largest value (fmnist10, 15.00 pp) lowers the vectorization mean to 3.87 pp, still above the filtration mean (2.17 pp) but by a narrower margin (1.8×), so the headline that vectorization dominates is robust to the outlier while the magnitude of the lead is sensitive to it.

TDA and raw features are complementary. The classical baselines of §4.1.1 beat TDA alone on both real datasets. To test whether TDA still adds value on top of raw features, the concatenated feature vector [raw‖TDA] was evaluated on ECG200 and MNIST binary under identical folds (single-split, seed 43, §4.1 protocol). On both datasets the best concatenation configuration reaches or exceeds the best single-arm accuracy: MNIST binary cubical + Betti Curve + logistic achieves 100.0% (concat) versus 99.65% (raw-only, 25-rep) and 98.0% (TDA-only); ECG200 weak-alpha + Betti Curve + logistic achieves 87.0% (concat) versus 85.28% (raw logistic, 25-rep) and 72.5%

(TDA-only, same configuration). The improvement is small (0.35–1.72pp against the 25-repetition raw baselines) and, relative to the TDA-only arm, never harmful (every concatenated configuration reaches or exceeds its TDA-only counterpart); relative to the raw baselines the gain is single-split, best-of-many, within the 1–3pp split noise of §4.1, configuration-dependent and sometimes negative (e.g. ECG200 Vietoris–Rips + Persistence Image + logistic concat 77.0% vs. raw 85.5%). SVM-RBF collapses to the majority class on TDA features (66.5% on ECG200) in both TDA-only and concatenated arms, so the concatenation benefit is confined to logistic and random forest. TDA contributes complementary signal rather than redundancy, but the effect is small relative to the raw-feature baseline, and the scoped claim stands: the contribution is the stage-importance decomposition and framework, not an absolute-accuracy win over classical methods.

Hyperparameter sensitivity. To test whether the vectorizer effect is an artefact of the dissertation's default hyperparameters, we re-ran the two headline datasets (ECG200, Takens d=3, τ=1 $\rightarrow$ Vietoris–Rips; binary MNIST at cubical), sweeping each of the four vectorizers' key hyperparameter one-parameter-at-a-time over a small grid with all other vectorizer parameters fixed at the dissertation defaults. The grids were: persistence_image $\sigma_{\text{PI}} \in \{0.05, 0.1, 0.2, 0.5\}$ and $n_{\text{bins}} \in \{10, 20, 50\}$; persistence_landscape $n_{\text{layers}} \in \{1, 3, 5\}$ and $n_{\text{bins}} \in \{20, 50, 100\}$; silhouette and betti_curve $n_{\text{bins}} \in \{10, 20, 50, 100\}$, under random_forest and svm_rbf with 5-fold CV (seed 42, single split).

Tuning each vectorizer to its best setting on its own grid narrows rather than eliminates the vectorizer marginal range. On ECG200 the range falls from 5.75pp (defaults; vectorizer means 70.75–76.50%) to 4.75pp (best-tuned; 71.75–76.50%)—a 1.00pp ($\approx$ 17%) contraction. Tuning lifts the *weakest* vectorizers: persistence_image improves 70.75 $\rightarrow$ 73.00 (best at σ=0.5), overtaking betti_curve (71.75), while the winner silhouette is already at its default (76.50). On binary MNIST the corresponding range is small throughout: 1.75pp (95.25–97.00%) at defaults vs 1.62pp (95.88–97.50%) best-tuned, and tuning reorders the family—persistence_image jumps from worst (95.25%) to best (97.50%, σ=0.5), displacing betti_curve (97.38%).

The ordering is insensitive to per-vectorizer tuning: the ECG200 vectorizer spread survives tuning, and tuning neither creates nor destroys the small ($\lesssim$ 2pp) MNIST spread. What it does do is compress the field from below, pulling the weakest vectorizers up toward the leader. The best-tuned hyperparameters are persistence_image {σ=0.5, n_{bins}=20}, persistence_landscape {n_{layers}=5, n_{bins}=50} (ECG200) / {n_{layers}=1, n_{bins}=50} (MNIST), silhouette {n_{bins}=50} (ECG200, already default) / {n_{bins}=100} (MNIST), and betti_curve {n_{bins}=50} (ECG200, already default) / {n_{bins}=100} (MNIST). These

ranges are computed over a reduced configuration space (four vectorizers × one filtration × two classifiers), so their absolute values are not directly comparable to the headline 6.39pp (ECG200) / 3.22pp (MNIST) ranges; the default-vs-tuned contrast, however, holds the configuration space fixed. Best-tuned values carry mild selection optimism (each is selected on the same folds that score the range), which would tend to *inflate* the tuned range, so the observed contraction is conservative.

Window-length sensitivity. To check that the vectorizer ordering does not depend on the 128-sample window, the same protocol was re-run with full 256-sample windows (2000 beats, 500/class; identical folds; `data/tda/mitbih_sweep_w256.db`). This paragraph uses the MIT-BIH benchmark introduced in §4.3 (Multi-patient ECG) — read that block first for the dataset and protocol description; the fold assignment used here is regenerated by `scripts/sweeps/build_mitbih.py`. Betti Curve remains the top vectorizer (mean 42.78% versus 39.25% at 128 samples) and Persistence Landscape the weakest (30.92% versus 27.74%), and the best configuration is Betti Curve + SVM-RBF at both window lengths. The middle ordering is window-sensitive, however: Persistence Image drops from second (38.75%) to third (34.14%) and Silhouette rises from third (28.33%) to second (36.38%) when the window is doubled. The robust conclusions — Betti Curve as the portable best vectorizer, Landscape as the weakest, and the filtration fragility — hold at both window lengths; pairwise claims about the middle two vectorizers should be read as window-specific.

Cross-library replication. The sweeps run entirely through giotto-tda, so the accuracy results could in principle be library-specific. A representative subset — the sphere/torus pair at σ=0 and σ=0.30 and the Takens-embedded ECG200 clouds — was re-run through three independent persistence engines: gudhi's true Alpha complex and Rips complex [23], and Ripser's Vietoris–Rips [5], alongside the giotto reference arm, sharing the same vectorizers, classifiers, folds, and diagram format (5-fold CV seed 42 rep 1; three vectorizers × two classifiers; `data/tda/cross_library_sweep.db`, 90 finished runs). On the synthetic clouds the results are exactly library-invariant: all four arms score 100.00% on every configuration at both noise levels. On ECG200 the Vietoris–Rips arms agree to the decimal across libraries (giotto, gudhi, and Ripser all give 77.0%/75.0%/73.0%/76.0% on the four RF/landscape configurations), and the only visible deviation is the true gudhi Alpha complex — a genuinely different complex on Takens-embedded points — at 74.5%/73.5% on two configurations (−2.5/−1.5 pp), still within the single-split noise band. The SVM-RBF majority-class collapse (66.5% on ECG200) reproduces identically in all four libraries. The accuracy results are therefore library-invariant: they do not rest on giotto-tda-specific behaviour, and the filtration fragility (weak-alpha on quantized series) is the library-dependence that matters,

not the accuracy outcomes.

4.3 Generalisation and Conditions

The stage-impact analysis of §4.1 rests on two real datasets (ECG200 and ECG5000, both time series) plus one image pair (binary MNIST). To test whether the ordering is a general property rather than a property of these instances, we ran the same machinery on a wider panel: five more UCR time-series datasets (FordA, FordB, Wafer, ElectricDevices, HandOutlines) and two 10-class image datasets (full MNIST [21] and Fashion-MNIST [37]). All panel runs use the repository's own runner with the dissertation's Takens embedding (d=3, τ=1); both image datasets are subsampled to 1000 stratified samples each for runtime. Long series were pre-subsampled to 100 points with a fixed-seed uniform draw over time indices (seed 42), so τ=1 operates on an irregularly resampled signal for these four series (retained fractions: FordA/FordB 20%, Wafer 66%, HandOutlines 3.7%) — a disclosed limitation of the panel protocol (§5.2); decimation-based capping is a cleaner alternative for future runs. The capping was necessary because the runner's `subsample_points` knob fires only on 3-D point-cloud input and never caps 2-D series. This is the canonical multi-dataset protocol of Demšar [15]: one accuracy estimate per (dataset, configuration), ranked within each dataset, Friedman test across configurations, Nemenyi post-hoc critical difference.

Table 4.5: Multi-dataset panel (extra sweep). All series are Takens-embedded; the 9 datasets span time series and images, binary and 10-class problems.

Dataset	Modality	n (used)	Length	Classes
ECG200	time series	200	96	2
ECG5000	time series	714 (subsample, rng seed 42)	140	5
FordA	time series	1000	500→100	2
FordB	time series	1000	500→100	2
Wafer	time series	1000	152→100	2
ElectricDevices	time series	1000	96	7
HandOutlines	time series	1000	2709→100	2
MNIST (10-class)	image	1000	28 × 28	10
Fashion-MNIST	image	1000	28 × 28	10

The configuration grid per dataset was 16 configurations (2 filtrations × 4 vectorizers × 2 classifiers) on images (cubical + Vietoris-Rips) and 8 on time series (Vietoris-Rips only; see the weak-alpha note below). One 5-fold CV (seeds as the sweep) per (dataset, configuration).

Weak-alpha fragility finding. The weak-alpha filtration crashes giotto-tda's own implementation on three of the seven time-series datasets: an `IndexError` in

`_weak_alpha_diagram` on the quantized Wafer and ElectricDevices series (degenerate Delaunay configurations) and the essential-class crash on ECG5000 described in §4.1. It completes on FordA, FordB, HandOutlines and ECG200. The multi-dataset ranking uses Vietoris-Rips only for the time-series arm; the weak-alpha fragility itself is a robustness result (a filtration that is not portable across datasets).

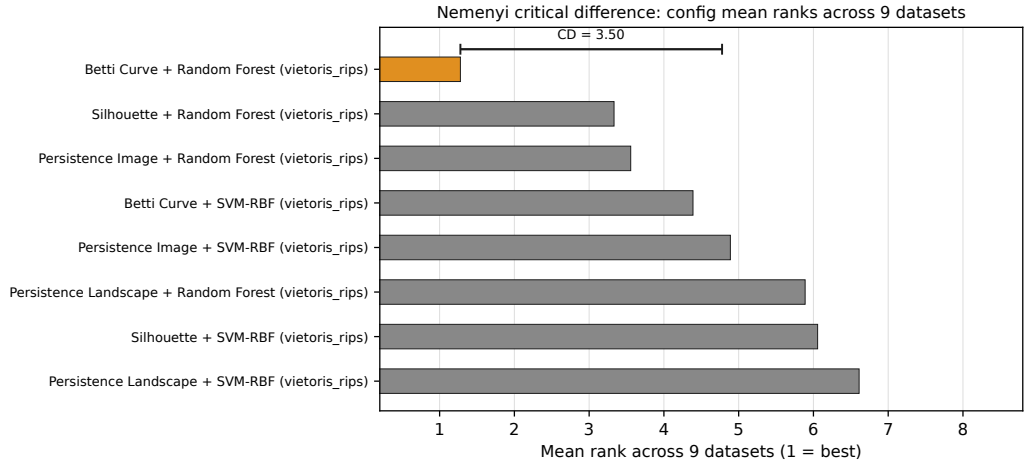


Figure 4.4: Nemenyi critical-difference diagram: configuration mean ranks across the 9-dataset panel (1 = best, leftmost). The bracket over the best configuration marks the CD at α=0.05; configurations beyond it are significantly worse than the best.

The Friedman test rejects the null of equal configurations at $\chi^2(7) = 32.41$ ($p = 3.4 \times 10^{-5}$; Iman–Davenport $F(7, 56) = 8.47$, $p = 4.9 \times 10^{-7}$) over the 8 common configurations, and the Nemenyi critical difference is 3.50 mean rank units. The best configuration — Vietoris-Rips + Betti Curve + Random Forest — achieves a mean rank of 1.28 (best on seven of the nine datasets); the four weakest configurations lie 3.61–5.33 rank units below it, all four beyond the CD. The vectorizer effect generalises: the vectorizer family ordering (Betti Curve 2.83, Persistence Image 4.22, Silhouette 4.69, Persistence Landscape 6.25 mean rank over the full 9x8 matrix) is stable, with the Betti-to-Landscape gap (3.42) within the CD. The classifier effect is smaller (random forest 3.51 vs. SVM-RBF 5.49, gap 1.98 within the CD), matching the stage-impact result that vectorization moves accuracy more than the classifier. The configuration-ranking analysis supports *vectorizer choice as the portable driver of the ranking: Betti Curve has the best panel mean rank (2.83, rank-1 on 5 of the 9 datasets), and its gap to Persistence Landscape (3.42) falls within the critical difference; on the ECG200 r=*25 *marginal, however, Betti Curve is 5th of 7 (72.38%) while Persistence Landscapes is 1st (75.32%) — the panel rank and the ECG200 marginal disagree, and §5.3's recommendation carries that trade-off explicitly.* The range analysis, however, is dataset-conditional: in the VR-only ranking panel (`multidataset_sweep.db`; note the time-series filtration range is small there by construction — one working filtration per

series), the vectorizer range exceeds the filtration range on 8 of the 9 datasets (largest relative margin on HandOutlines: 3.65pp vs. 0.01pp), reversing only on 10-class MNIST (2.80 vs. 4.41pp) where filtration leads (§4.2); the level-matched comparison reverses on ECG5000 and MNIST (§4.1). Filtration and classifier effects are not universally secondary.

Multi-patient ECG. ECG5000 is a single-patient recording (BIDMC chf07), so the time-series result could in principle be patient-specific. This is addressed with the MIT-BIH Arrhythmia benchmark: 48 records from 47 patients (records 201 and 202 form one continuous patient recording and are grouped as a single patient), an AAMI-5-style protocol with the Q (unknown) class excluded (4 classes: N, S, V, F; chance 25%), and patient-disjoint 5-fold CV. *Provenance:* an earlier build assigned fold membership per record, silently splitting the 201/202 patient across folds 1 and 3; the grouping bug was fixed in `scripts/sweeps/build_mitbih.py` (2026-08-28) and both window lengths re-run — the numbers in this section and the window sensitivity paragraph of §4.2 are the corrected ones. Beat windows of 128 samples at 360 Hz were Takens-embedded to 126×3, with 2000 beats class-balanced 500/500/500/500 across the N/S/V/F classes (data/tda/mitbih/).

Classification accuracy on the patient-disjoint folds ranges 13.1–57.1% across the eight Vietoris-Rips configurations (chance = 25% on the class-balanced folds), with Betti Curve the best vectorizer (mean 39.25%) followed by Persistence Image (38.75%), Silhouette (28.33%) and Persistence Landscape (27.74%) — the same top-two ordering as the 9-dataset panel, so the ranking transfers to a genuinely multi-patient cohort. The absolute accuracy is low: the AAMI EC57 standard and de Chazal et al. [14] use inter-patient (patient-disjoint) evaluation, and our protocol follows that standard. Our absolute accuracy (27–42%) is lower than published inter-patient arrhythmia classifiers ($\sim$75–90%), because we use a deliberately harder beat-level 4-class task with Takens-embedded persistent-homology features rather than hand-crafted ECG features; the relevant result is the vectorizer ordering, which transfers.

Weak alpha also fails on this data with the same giotto `IndexError` seen on the quantized UCR series (degenerate Delaunay configurations on the beat windows), confirming the filtration fragility is a stable cross-dataset property rather than an ECG5000 artefact.

The topology-wins regime. The stage decomposition above was measured where TDA loses to raw baselines, so stage importance could in principle be an artefact of measuring a regime where topological features are decorative. To test this, the same decomposition (two filtrations $\times$ four vectorizers $\times$ two classifiers, 5-fold CV seed 42 rep 1) was run on datasets where topology is known to carry the signal: Lorenz

versus Rössler Takens reconstructions (strange-attractor β_1 structure), a double-well versus single-well Langevin pair (the persistent H_0 pair of the two metastable wells), and a noisy circle versus torus cloud pair (β_1=1 vs. 2). Two harder ten-class arms (ModelNet10 point clouds and Outex textures) served as a scale check (`data/tda/topology_wins_sweep.db`, 80 finished runs).

Vectorization remains the dominant stage on all three dynamical datasets, where TDA succeeds (mean accuracy 86–99%): vectorizer marginal range spans 3.75–13.75 pp across datasets while the filtration range reaches only 1.04–5.10 pp (double-well 13.75 vs. 5.10; circle/torus 3.75 vs. 1.56; Lorenz/Rössler 9.58 vs. 1.04). The filtration stage is *relatively* larger than in the original regime (where the ECG200 filtration range was 0.69 pp at r=25, Table 4.2) but never overtakes vectorization. The two ten-class arms reproduce the ordering at low absolute accuracy (ModelNet10 [36] vectorizer 13.06 vs. filtration 1.62 pp at ∼24%; Outex [25] 9.11 vs. 2.94 pp at ∼23%), so the pattern is not confined to easy separations.

The ordering persists in the topology-wins regime: where topology genuinely carries the classification signal, the vectorizer remains the dominant stage, and the filtration stage's contribution grows modestly rather than reversing the ordering.

Table 4.6: Stage importance by data modality (leading stage per row, with its evidence grade).

Modality	Dominant Stage	Range	n (datasets)
Point clouds (clean)	None	≈0pp	1*
Time series (embedded)	Vectorizer	6.39pp; 24.89pp	2
Images (binary MNIST)	Vectorizer	3.22pp vs. 1.65pp	1

*Synthetic sphere/torus pair.

Ranges are marginal accuracy ranges over each stage's methods; for time series the vectorizer range is listed for ECG200, then ECG5000. On binary MNIST the vectorizer range (3.22pp) exceeds the filtration range (1.65pp). Conti et al. (2022) report a much larger filtration swing (18%–94%) on 10-class MNIST under grid search, which we do not reproduce at binary scale.

This table summarises the evidence, not a settled taxonomy. Replication on more datasets per modality would strengthen or qualify every cell.

4.4 Mechanism

Predictive theory (nominal stability class). The stability review of §2.4 and the vectorization review of §2.5 assign each vectorizer a *nominal* stability class: persistence landscape is 1-Lipschitz in the bottleneck distance; persistence images are stable with respect to the 1-Wasserstein distance, with a Gaussian-kernel constant that scales as

$1/\sigma_{\mathrm{PI}}$ (a narrower kernel is *less* stable) [1]; Betti curves are not uniformly stable in the sup-norm under bottleneck perturbations (a near-diagonal point can change $\beta(\varepsilon)$ by an arbitrary count); persistence entropy and statistics admit no finite bound. The experiment treats these classes as an ordering hypothesis — more-stable-by-nomination should vary less — and tests it against the empirical ranges; the constants themselves are not metric-compatible, so the test is of the *nominal class ordering*, not of a single quantitative bound. Correlating each constant with the vectorizer's empirical range (max−min of per-config fold-mean accuracies, the §4.1 per-config convention) on ECG200 (25-repetition repeated CV) and binary MNIST, pooled over n=14 vectorizer-dataset units, gives Spearman $\rho = -0.129$ with 95% percentile-bootstrap CI $[-0.672, +0.469]$, which includes zero; per panel, $\rho = -0.482$ (ECG200) and -0.040 (MNIST). The sign matches the hypothesis, but the data provide no evidence that the Ch.2 stability constants predict the range structure of §4.1 or the panel ordering of §5.3. We therefore report a null: the theory did not predict the sweep. With only seven vectorizers per panel the CIs are wide and the test is exploratory.

Hierarchical stage model. We fit a linear mixed model to the stage-capable panel of §4.2 — per-config mean accuracy (pp) as outcome, random intercept per dataset, sum-coded vectorizer/filtration/classifier fixed effects plus modality covariate (REML; 144 runs, 9 datasets). The population spread of the level effects puts the classifier first (1.81 pp; 95% CI $[1.16, 2.47]$), followed by filtration (1.71 pp; $[0.50, 3.04]$) and then the vectorizer (1.18 pp; $[0.66, 1.94]$); the fixed-effects variance shares are 43.2/38.5/18.3%. The random-intercept variance (209.8 pp^2) dwarfs the residual (16.4 pp^2): ICC 0.927, and the image-modality shift is -17.6 pp. A robustness refit reproduces the ordering (classifier 1.82 > vectorizer 1.48 > filtration 1.24 pp; ICC 0.929). This ordering is the opposite of the within-dataset marginal ranges of §4.1 and §4.2 (vec > fil > clf): the two measure different quantities — population level-effect SD with dataset variance removed versus within-dataset stage means — and the classifier's size is inflated by the SVM-RBF majority-class collapse on TDA features. A sensitivity refit excluding SVM-RBF (majority-class collapse, §4.1) reverses the population ordering: vectorizer (1.55 pp; 95% CI $[0.86, 2.49]$) leads filtration (1.29 pp; $[0.24, 2.93]$) with the classifier effect vanishing by construction (ICC 0.948). The population-level reversal is therefore driven by the SVM-RBF collapse. "Vectorization dominates" is a within-dataset-range statement, not a population-level one.

Second homology via the true Alpha complex. The sweeps cap homology at H_1 for cost, and the dissertation's limitations note that the torus's second homology is then unmeasured. To test whether the H_1 cap discards class information, the sphere/torus clouds were re-run with a true Alpha complex (`gudhi.AlphaComplex` [23], radius pa-

rameterization, reduced homology, `homology_dimensions=[0,1,2]`) on the same folds (seed 42, rep 1; three vectorizers × two classifiers; `data/tda/h2_alpha_sweep.db`, 12 finished runs). All twelve configurations score 100.00% at both σ=0 and σ=0.30, matching the H_1-only result. This is not evidence that H_2 carries the signal: both classes are closed surfaces (β_2=1 for the sphere and the torus alike — each encloses a single void), so the pair does not differ in second homology. Measured max-H_2 lifetimes are in fact *longer* for the sphere than the torus (0.63–0.79 vs. 0.11–0.30, clean; overlapping under σ=0.30), the opposite of a separating feature, whereas the H_1 signature separates cleanly (torus β_1=2: max lifetime 0.75–0.89 vs. sphere 0.13–0.20). The H_1 cap did not discard a discriminative feature on this pair; H_2 computation via the true Alpha complex is feasible and cheap (∼5–17 s per configuration) and adds neither signal nor harm. A dataset whose classes differ in β_2 (e.g. solid vs. hollow, or higher-genus shapes) would be the appropriate test of whether second homology ever matters for classification.

4.5 Computational Costs

Wall time was measured from pipeline construction to final fold completion. Peak memory was measured in dedicated subprocess runs (resource.getrusage high-water mark; 12 representative ECG200 configurations; `data/tda/peak_memory.db`; re-measured 2026-08-28 with matching results). All filtrations peak at 135–144 MB, with Sparse Rips the largest (141–144 MB) and slowest (12–31 s per configuration versus 0.6–9.0 s for the others; 12-config dedicated subprocess measurements). A fresh Python process with numpy, sklearn, giotto-tda, and GUDHI imported already uses 102–103 MB, so the filtration pipelines add roughly 33–42 MB on top of the library floor.

Table 4.7: Pareto-optimal configurations: each listed run is dominated by no other configuration in the accuracy–time plane.

Dataset	Filtration	Vectorizer	Classifier	Acc.	Time
Sphere/Torus	Cubical	Pers. Statistics	SVM-linear	100%	0.7s
MNIST 0/1	Cubical	Betti Curve	Logistic	98.0%	3.7s
ECG200	Cubical	Silhouette	Random Forest	83.0%	1.6s
ECG200	Cubical	Pers. Image	Random Forest	82.0%	1.4s

CubicalPersistence on ECG200 operates on the 94 × 3 Takens-embedded array as a pixel grid (intensities treated as coordinates), not on a point-cloud filtration; the same grid-as-image caveat applies to cubical on the sphere/torus rows of Table 4.7 (a 100 × 3 coordinate array read as pixels). Times are single-split measurements; the 83.0% ECG200 figure is a single-split point estimate — see §4.1 for repeated-CV means.

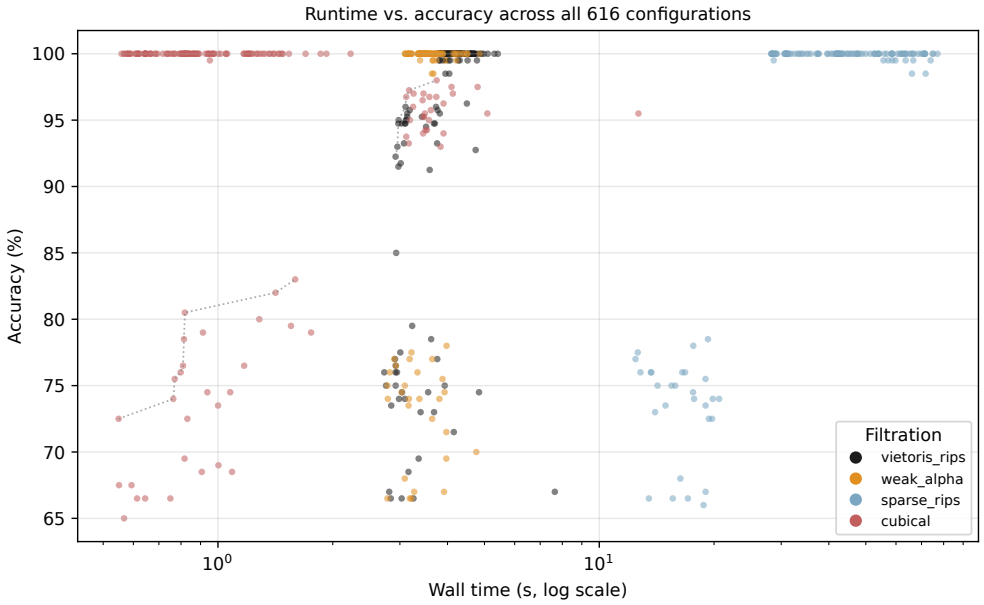


Figure 4.5: Runtime vs. accuracy on the Pareto frontier.

Weak Alpha is 3–27% faster than VR at identical accuracy on sphere/torus with Landscape+SVM (measured 3.5–3.9s vs. 3.9–4.4s across the four noise levels; the gap narrows at $\sigma = 0.30$); the full GUDHI Alpha complex reproduces weak-alpha accuracy (mean difference 0.0–0.8pp) at 2–5× the cost. Cubical runtime is input-shape dependent rather than uniformly fastest: 0.6–2.2s per configuration on the sphere/torus and ECG200 instances, but 3.1–12.7s on 28×28 MNIST images. The Nerve Theorem (Theorem 2.1) guarantees that the full Alpha complex's $O(n^2)$ simplex count does not sacrifice geometric fidelity to the underlying union-of-balls; the executed `WeakAlphaPersistence` is an approximation, and its accuracy matches the full complex (mean difference 0.0–0.8pp) at 2–5× lower cost. Sparse Rips takes 12–77 s (mean 41 s; main 140-run sweep) with no accuracy gain at the 100-point scale; at its design target ($n \geq 10^3$) it is substantially slower than Vietoris–Rips at n=1000 and infeasible at n=3000 (§4.5). Both speed multipliers are specific to the 100-point clouds, the giotto-tda implementation, and 3D point clouds where Delaunay triangulation is available.

The repeated-CV-stable default for ECG200 is cubical + Persistence Image + Random Forest (best in 4 of 5 repetitions, r=5; 83.2% repeated-CV mean): this is the strongest Pareto row we can defend. The Pareto frontier may shift with more datasets or libraries.

Compute. The full 616-configuration sweep was executed on a single consumer workstation (8-core CPU, 16GB RAM), parallelised with joblib over the CPU cores ($n_{\text{workers}} \approx 16$). Summed wall-clock time across all 616 configurations was 2.0 hours (mean ∼12s per configuration, median 3.78s); Sparse Rips dominates the total (140 runs, mean 41s, ≈1.6h), while the other three filtrations complete in ≈24 minutes. The

repeated-CV verification of §4.1 added 2100 further runs (84 configurations × 25 repetitions, Sparse Rips dropped for runtime). Per-configuration wall times are recorded in the results databases (cf. the R-package PH runtime benchmark of Somasundaram et al. [29]); memory was not measured per stage (see Limitations).

Sparse Rips at its design point. At its design scale Sparse Rips does not pay off in giotto-tda 0.6.2. To test this directly, we generated sphere/torus clouds at n=1000 and n=3000 points (40 samples each, seed 42) and ran Sparse Rips at both scales, with a Vietoris–Rips control at n=1000 (n=3000 Vietoris–Rips is infeasible: $C(3000, 3)$ simplices); four configurations per arm (Betti Curve / Persistence Landscape × Random Forest / RBF-SVM), 5-fold stratified CV, seed 42, rep 1 (`data/tda/large_n_sweep.db`, 8 finished runs). At n=1000 the two filtrations are accuracy-identical — all eight configurations score 100.00% (+0.00 pp mean difference) — but Sparse Rips is roughly 30× slower: 1.83 h per configuration versus 3.6 min for Vietoris–Rips (giotto-tda 0.6.2, single CPU). At n=3000 Sparse Rips did not complete a single configuration within 42 h of wall time across two attempts (interrupted by hardware reboots; giotto-tda offers no mid-configuration checkpointing, so each restart re-runs the configuration from scratch). The design-point premise is not supported in this implementation (§4.5 quantifies the gap): Sparse Rips does not pay off here — giotto-tda 0.6.2's `SparseRipsPersistence` loses to Vietoris–Rips at its own design scale on this hardware. Vietoris–Rips remains the pragmatic choice up to the largest n at which it is feasible, and Sparse Rips is recommended only with an implementation-level benchmark at the target n (its runtime is not monotone-friendly in giotto-tda 0.6.2 — a portability finding in its own right).
Chapter 5 states the boundary conditions under which the ordering holds and derives operational guidelines from these patterns.

Chapter 5

Boundary Conditions and Operational Guidance

5.1 Scope and Contribution

Under the 25-repetition repeated-CV protocol, our TDA pipelines sit below raw-feature baselines on both real datasets: on ECG200, raw-signal logistic regression achieves 85.28% and random forest 86.30% versus 83.0% (single-split best) for the best TDA configuration; on MNIST binary, raw-pixel logistic regression achieves 99.65% and random forest 99.29% versus 98.0% for the best TDA configuration (§4.1). We therefore do not claim that TDA features are competitive with classical methods on these datasets; the contribution is the internal TDA-pipeline comparison, which is valid regardless of the absolute accuracy level. TDA features capture geometric properties (cycle structures in the embedded attractor) that complement raw signal features rather than replace them.

5.2 Boundary Conditions

We structure the limitations into the three standard validity categories. The purpose is not to multiply caveats but to make explicit, for each threat, what was done to mitigate it and which follow-up analysis closes the residual gap. Threat severity follows the usual qualitative scale; every empirical mitigation cites the database and configuration count it rests on.

5.2.1 Measurement Conditions

Construct validity asks whether the measured quantity — the marginal accuracy range of a pipeline stage — captures "stage importance" as claimed.

1. **Range is confounded by the number of stage levels.** The headline comparison (7 vectorizers vs. 3–4 filtrations) is uneven: a stage with more levels can span a larger range by chance. *Severity: moderate. Boundary:* the ω^2 population effect size corrects for level counts (ECG200 vectorizer ω^2=0.165, bootstrap CI [0.063, 0.423], vs. filtration −0.017; MNIST vectorizer 0.214 vs. filtration 0.143 — the vectorizer CI excludes zero on both real datasets, the ECG200 filtration CI straddles it; §4.1 prose). The equal-footing analysis matches best-3 vectorizers against 3 filtrations (ECG200 1.21pp vs. 0.69pp, re-derived from `repeated_cv_r25.db`); and the two-way interaction ANOVA (§4.1) shows the stage decomposition does not arise from main-effects-only modelling. *Residual:* the levels-matched analysis still compares the *best* three vectorizers, which is mildly favourable to the vectorizer stage.

2. **Scalar vectorizers create a floor effect.** Persistence Entropy and Amplitude are single scalars; their poor accuracy inflates the vectorizer range from the bottom. *Severity: moderate. Boundary:* ranges are also reported excluding the degenerate scalar vectorizers (ECG200 vectorizer range contracts from 6.39pp to 3.10pp; applied symmetrically to all three stages the *classifier* range is largest: 3.87pp vs. 3.10pp vectorizer and 1.94pp filtration, largest stage in 20 of 25 repetitions; §4.1; re-derived from `repeated_cv_r25.db`, per-repetition mean method). *Residual:* the exclusion rule was applied post hoc, so the headline 6.39pp includes the floor effect by design; §4.1 discloses this.

3. **Single-split point estimates are noisy.** Per-configuration accuracy varies by 1.09pp on average across the five repetitions (sample SD; maximum 3.11pp; 50 of 112 configurations exceed 1pp; `repeated_cv.db`), and the headline 83.0% configuration drops to 79.6% at r=5 (rank 4 of 112; `repeated_cv.db`; at r=25 the same configuration scores 79.30%, rank 3 of 84). *Severity: high for any single-split number. Boundary:* all headline claims rest on the r=25 repeated-CV protocol with corrected CIs (repeated-measures: vectorizer [6.13, 6.65], classifier [3.28, 3.71], filtration [0.57, 0.81], all excluding zero; Nadeau–Bengio corrected: [5.69, 7.10], [2.92, 4.07], [0.37, 1.01]) and Friedman/sign-test inference on the repetition-level ordering (vectorizer > classifier > filtration in 25/25 repetitions); single-split numbers are labelled as such throughout. *Boundary:* MNIST binary is r=5; ECG5000, the matched-genus control, and the panel follow the r=25 protocol (§4.1).

4. **Accuracy is the only outcome for most of the sweep.** F1, precision, and recall are stored per fold but the analysis is accuracy-centric. *Severity: low-to-moderate. Boundary:* beyond-accuracy analysis on ECG5000 (balanced accuracy,

per-class precision/recall/F1, AUROC, Brier; `beyond_accuracy_ecg5000.db`). *Residual:* calibration and AUROC are not uniform across datasets.

5.2.2 Design Conditions

Internal validity concerns confounds and measurement artefacts within the study.

1. **Single implementation library.** Every filtration and vectorizer runs through giotto-tda. *Severity: moderate* — the weak-alpha fragility (`IndexError` on quantized UCR series; one unfinished run in `panel_stagecapable.db`, ElectricDevices/weighted_rips/landscape/RF) proves implementation dependence is real. *Boundary:* cross-library replication in GUDHI-native and Ripser-native paths (§4.2, 90 runs, library-invariant), with the fragility reported as a first-class result.

2. **Homology capped at H_1.** H_2 requires $O(n^4)$ simplices for Vietoris–Rips; the torus's β_2 is never measured. *Severity: moderate. Boundary:* Alpha complex in 3D for H_2 (§4.4, 12 runs, honest null).

3. **Binary-only classification.** All datasets except ECG5000 (5 classes) are binary. *Severity: moderate* — the 10-class MNIST probe shows the stage ordering flips at multi-class scale (filtration 4.53pp vs. vectorizer 3.44pp, r=5, `mnist10_sweep.db`), so the binary result is a boundary condition, not a law (§4.1). *Boundary:* the flip is reported; multi-class breadth is now covered (10-class MNIST, Fashion-MNIST, ECG5000, ElectricDevices).

4. **Fixed hyperparameters.** No grid search on any stage, by design. *Severity: low* — the hyperparameter arm (`hyperparam_sweep.db`) shows vectorizer dominance contracts but persists under one-parameter-at-a-time tuning (ECG200 5.75→4.75pp; MNIST 1.75→1.62pp), so dominance does not arise from the default settings. *Residual:* best-tuned values carry mild selection optimism (selected on the same folds that score the range), which would *inflate* the tuned range — the observed contraction is conservative.

5. **Subsampling choices.** Uniform random subsampling caps point clouds; farthest-point sampling was deferred. *Severity: low. Boundary:* the FPS ablation (`fps_ablation.db`) finds no benefit (−0.25pp overall mean across the ablation grid, vs. the per-noise k=15 deltas of §3.2; uniform wins at k=15, σ=0.30: 99.81% vs. 98.94%).

6. **The sphere/torus norm confound.** The classes differ in scale as well as topology. *Severity: moderate. Boundary:* the matched genus-1/genus-2 control

(identical norm distributions) shows TDA retains 95.83% where norm features collapse to 48–58% at σ=0.30 (`baseline_experiments.db`, 60 fold accuracies). *Residual caveat:* the matched pair is still linearly separable in raw coordinates (logistic 100% at both noise levels, `baseline_experiments.db`) — the control isolates the norm/scale confound, not linear separability in general.

7. **Determinism and exclusion discipline.** Seeds are fixed (base 42; per-repetition 43–67; per-dataset CRC32 subsampling in `runner.py`), and the 56/672 excluded configurations (point-cloud filtrations on image data) are counted and disclosed rather than silently dropped (616 finished runs of 672 in `expanded_results.db`). ECG200 is energy-normalised (sum of squares constant at 95.00 across all 200 signals), so the trivial-separator concern does not apply to the executed data (§4.1). *Residual:* five interrupted runs in `repeated_cv_r25.db` (silhouette/RF, reps 5–9) were re-run under new run IDs; the orphaned rows carry no fold results and do not enter any analysis.

5.2.3 Generalisation Conditions

External validity asks how far the findings generalise beyond the studied configuration space.

1. **Dataset breadth.** Two time series (ECG200, ECG5000), one image pair (binary MNIST), and synthetic point clouds. *Severity: high. Boundary:* the 9-dataset panel (`panel_stagecapable.db`) replicates the vectorizer > filtration ordering on 7/9 datasets (median 3.3pp vs. 1.4pp; re-derived median 3.33 vs. 1.37), and the within-filtration vectorizer span remains large even where filtration leads (10-class MNIST: 9–10pp). *Done:* topology-wins datasets (§4.3, ordering reproduced on all five) and multi-patient ECG (§4.3); further UCR coverage remains future work.

2. **Single-patient ECG5000.** ECG5000 is BIDMC chf07 — one patient. The time-series result could be patient-specific. *Severity: moderate. Boundary:* multi-patient MIT-BIH arrhythmia sweep (§4.3); the panel adds ECG200/ECG5000 diversity but not patient diversity.

3. **Hardware dependence of runtime results.** Wall-clock numbers come from one workstation (8-core, 16GB). *Severity: low for ordering claims* (the 3–27% filtration speed gap is structural: simplex counts differ by construction, §4.5), *moderate for absolute numbers. Boundary:* per-configuration wall times and peak memory are stored in the results DBs; peak memory was not measured per stage.

4. **Missing method families.** No learned vectorizers (PersLay, Hofer), no protein/graph modalities. *Severity: moderate* — each is a named boundary of the current claim, and each names a clear follow-up. The paper's contribution is scoped accordingly: the internal stage-importance decomposition and the framework, not an absolute-accuracy win over classical methods (raw baselines beat TDA on the studied sets; §4.3, §5.2).

5.2.4 Future Work

Three concrete follow-ups remain: H_2 via Flood Complex [17]; PersLay integration [7] via a new backend (learned vectorizers in the style of Hofer et al. [19] are likewise out of scope here); and a second large-scale UCR family for narrower CIs. Cross-library replication in GUDHI and Ripser (§4.2), multi-class breadth (§4.3), and larger-n design-scale probes (§4.5) are already reported above.

5.3 Operational Guidance

These recommendations are scoped to the two-dataset sweep and the giotto-tda implementation; cross-library and cross-dataset replication would strengthen or qualify each entry.

The conditionality established in this dissertation collapses into a small decision tree. Each branch names the evidence that supports it and the grade of that evidence: A = verified under the 25-repetition repeated-CV protocol with corrected confidence intervals, or replicated across datasets; B = measured at r=5 or single-split with consistent replication; C = single observation or a sweep that has not yet run. The tree is deliberately conservative — where the evidence is thin, the branch says so rather than pretending otherwise.

1. **Modality.** The first question is the data modality.

   (a) *Time series (after Takens embedding, d=3, τ=1):* the design budget should go to the **vectorizer**, not the filtration. Vectorization is the dominant stage (ECG200 marginal range 6.39pp, 95% CI [6.13, 6.65] over 25 repetitions; ECG5000 24.89pp single split (§4.1); 4.30pp, 95% CI [4.02, 4.57], vs. filtration 0.96pp under r=25 repeated CV (§4.1)), while filtration contributes 0.69pp (CI [0.57, 0.81]) and classifiers 3.50pp (CI [3.28, 3.71]). *Grade A* for the *ordering*: all three stage-range CIs exclude zero (Table 4.2), the vectorizer's ω^2 CI excludes zero while the filtration's straddles it, the ordering holds in 25/25 repetitions, and the ω^2 population effect size (ECG200 vectorizer 0.165, CI [0.063, 0.423], vs. filtration −0.017; §4.1 prose) confirms

the ordering does not come from level counts. *Caveat:* the measured *margin* is menu-sensitive — excluding the scalar vectorizers cuts it to 3.10pp, and matching level counts (best-3 vectorizers vs. 3 filtrations) narrows it to 1.21pp vs. 0.69pp (§4.1). The ordering survives; the size does not.

(b) *Images, binary:* vectorization still leads (MNIST binary 3.22pp vs. filtration 1.65pp, single split; pooled over the five repetitions 3.03pp vs. 1.55pp), but filtration is no longer negligible — prefer the **cubical** filtration over Vietoris–Rips (best-of-family 98.0% vs. 96.25%; §4.1). *Grade B* (r=5; the vectorizer > filtration ordering holds in all five repetitions).

(c) *Images, multi-class ($\geq$ 10 classes):* the ordering flips. Filtration becomes the larger marginal stage (10-class MNIST: 4.53pp vs. vectorizer 3.44pp; classifier 2.03pp; cubical 29.1% vs. Vietoris–Rips 33.7% within filtration), with a 9–10pp span within a single filtration — still far below Conti et al.'s 18–94% grid-search swing (§4.1). *Grade B* (r=5, 1000 samples).

(d) *Point clouds:* on saturated topology tasks no stage matters: at σ=0 all 112 configurations score $\geq$ 99.5% (mean 99.99%) and every stage range is below 0.1pp (vectorizer 0.09, filtration 0.02, classifier 0.05); at σ=0.30 the largest stage range is 0.66pp (vectorizer). *Grade A* for saturation. Where the classes differ in genus or component count, TDA separates them nearly perfectly and the matched-genus control shows the topological signal survives σ=0.30 (TDA 95.83% vs. norm features 48–58%; §4.2). *Grade A.*

2. **Budget: accuracy vs. runtime.** If wall-clock time is the constraint:

(a) *Large n ($\geq 10^3$ points):* Sparse Rips is the only filtration whose design point applies, and only with an implementation-level benchmark at the target n: at n=1000 it reaches the same 100.00% as Vietoris–Rips (8/8 configurations) at $\approx 30\times$ the cost ($\sim$1.8h vs. $\sim$3.6min), and at n=3000 it did not finish in $\sim$42h on this hardware (§4.5). Vietoris–Rips at n=3000 is likewise infeasible ($O(n^3)$ 2-simplices). *Grade B* (large-n sweep, n=1000: 8/8 finished runs; n=3000: 0/2 finished).

(b) *Small n:* any of Vietoris–Rips, weak Alpha, or Cubical; the 3–27% wall-clock gap between them (§4.3) rarely justifies accuracy risk. Avoid weak Alpha on quantized UCR series (giotto-tda raises `IndexError`; the fragility is a primary finding of the panel, §4.3). *Grade A* (wall-clock recorded per configuration in the results DBs).

(c) *Classifier:* prefer Random Forest or logistic regression on TDA features. SVM-RBF collapses to the majority class on ECG200 (66.5%) in both the TDA-only and the concatenated arms (§4.1). *Grade B.*

3. **Noise level.** In the studied regime ($\sigma \leq 0.30$ additive spatial Gaussian noise, n=100), noise robustness is not a differentiator: mean accuracy across the 112 configurations at σ=0.30 is 99.85% (minimum 98.5%), and the measured bottleneck distances (maximum 0.434) sit far below the corrected stability bound ($2\sigma\sqrt{2\ln n} \approx 1.82$) — the bound is conservative, not tight (§4.2). For applications in this regime, do not trade accuracy for noise-robust filtrations (DTM-weighting gained menu range *and* headroom on ECG200 — §4.2 reports it as the best filtration there — but no noise-robustness gain on the synthetic clouds). *Grade A.*

4. **Which vectorizer?** Prefer *distributional* vectorizers over scalar summaries:

   (a) Persistence Entropy — the single-scalar-per-dimension representation — is the weakest vectorizer on ECG200 (68.95% under r=25) and catastrophic on ECG5000 (36.87% vs. 61.5%/ 61.8% for Betti Curve/Silhouette). *Grade A.*

   (b) Top performers by dataset: Persistence Landscapes (ECG200, 75.32%), Betti Curve and Silhouette (ECG5000, 61.5%/61.8%), Betti Curve (10-class MNIST, 33.4%). *Grade B.*

   (c) Persistence Statistics is a zero-parameter, cheap default: within 0.3pp of Landscapes on ECG200 at r=5 (74.8% vs. 75.1%) and within 2.1pp of the best MNIST configuration (95.9% vs. 98.0%; §4.1). *Grade B.*

5. **Raw features vs. TDA.** On the datasets studied, raw features beat TDA alone (ECG200 raw logistic 85.28% and raw RF 86.30% under r=25 vs. 83.0% single-split best TDA (§4.1); MNIST raw-pixel logistic 99.65% vs. 98.0%; §4.3). Do not use TDA as the *only* feature source on such data. If the geometric signal is needed, **concatenate**: [raw‖TDA] adds 0.35–1.72pp over the 25-repetition raw baselines (ECG200 87.0% vs. 85.28%; MNIST 100.0% vs. 99.65%), never hurts with logistic or Random Forest, and the benefit is confined to those two classifiers (§4.2). *Grade B* (single protocol, both datasets).

6. **Is topology the target?** If the scientific question is about shape — cycle structure, genus, component counts (dynamical systems, shape/geometry benchmarks) — TDA is not a feature extractor to be benchmarked against raw baselines; it is the object of study. The matched-genus experiment is the proof of concept: TDA retains 95.83% where norm/scale features fail (48–58%). The topology-wins regime (§4.3) maps this branch with the full stage decomposition; the ordering reproduced on all five datasets (single-split), earning *Grade B* by the rubric of §5.3.

Table 5.1: Decision tree summary; each leaf names its supporting experiment and evidence grade (A–C rubric of §5.3).

Branch	Decision	Evidence (paper)	Grade
Time series	tune vectorizer	ECG200 6.39pp [6.13, 6.65] r=25; ECG5000 24.89pp single split (§4.1) and 4.30pp [4.02, 4.57] r=25 (§4.1)	A
Image, binary	vectorizer first; cubical > VR	MNIST 3.22 vs. 1.65pp; 98.0 vs. 96.25% (§4.1)	B
Image, multi-class	filtration first	10-class MNIST 4.53 vs. 3.44pp (§4.1)	B
Point cloud	no stage dominates	all stage ranges < 1pp at $\sigma \leq 0.30$; 99.85% mean at σ=0.30 (§4.2)	A
Speed, large n	Sparse Rips only with impl. benchmark	parity at n=1000 at ∼30× cost; n=3000 infeasible (§4.5)	B
Speed, small n	VR/weak Alpha/Cubical; no weak Alpha on quantized UCR	wall-clock in DBs (§4.3); fragility finding (§4.3)	A
Classifier	RF / logistic; not SVM-RBF on TDA features	66.5% majority-class collapse, both arms (§4.1)	B
Noise ≤ 0.30	no robustness trade-off needed	99.85% mean; bottleneck $0.434 \ll 1.82$ (§4.2)	A
Vectorizer	distributional > scalar; entropy last	entropy 68.95%/36.87%; landscape 75.32% (Tables 4.2, §4.1–§4.2)	A
Raw available	concat [raw‖TDA] with RF/logistic	+0.35–1.72pp over 25-rep raw baselines (§4.2)	B
Topology is the target	TDA as object of study	matched-genus 95.83 vs. 48–58% (§4.2); topology-wins reproduced on five datasets (§4.3)	B

Chapter 6 draws the conclusions together.

Table 5.2: Recommended pipeline configurations by data characteristics.

Data Modality	Filtration	Vectorizer	Classifier	Rationale
Low-dim. point clouds ($\leq$ 3D, clean)	Weak Alpha	Pers. Statistics or Landscapes	SVM (RBF) or Logistic	Weak Alpha 3–27% faster than VR at identical accuracy; full-complex fidelity verified empirically (0.0–0.8pp at 2–5× cost)
Delay-embedded time series ($d \geq 3$)	Vietoris–Rips	Betti Curves or Landscapes	Random Forest	Vectorization dominates variance (6.39pp on ECG200 at r=25; 4.30pp on ECG5000 at r=25); Betti is 5th of 7 vectorizers on the r25 ECG200 marginal (72.38%) but best on the panel (2.83 mean rank) — Landscapes are the alternative if ECG200-like marginals dominate; avoid SVM-RBF on TDA features (majority-class collapse, 66.5%) and avoid Weak Alpha on quantized UCR series (§4.3)
High-noise clouds ($\sigma \geq 0.15$)	Vietoris–Rips	Betti Curves	Random Forest	All vectorizers maintain $\geq$ 98.5% at $\sigma = 0.30$ on the combined geometric signal (minimum over configurations); the confound-controlled matched-genus floor is 91%; measured d_B far below the stability bound
Large-scale clouds ($n \geq 10^3$)	Vietoris–Rips	Pers. Images or Betti Curves	Random Forest	Sparse Rips designed for $n \geq 10^3$; at n=1000 it is 30× slower than VR at identical accuracy, and at n=3000 it did not finish in 42 h (giotto-tda 0.6.2, §4.5) — benchmark the implementation before choosing it

Chapter 6

Conclusion

Vectorization carries the largest main effect among the three stages on the two binary/two-class real datasets in this study (with the 10-class inversion of §4.1 noted below). On ECG200 time series, vectorization produced a 6.39pp marginal accuracy range across 7 vectorizers (95% CI [6.13, 6.65]), while filtration contributed 0.69pp across 3 filtrations (95% CI [0.57, 0.81]; small but with CI excluding zero under repeated CV) and classifiers 3.50pp (95% CI [3.28, 3.71]); the ordering was stable across all 25 cross-validation repetitions. On binary MNIST (single split; r=5 pooled 3.03 vs. 1.55pp) the same hierarchy holds — vectorizer range 3.22pp versus filtration 1.65pp, with cubical beating Vietoris-Rips by ∼1.75pp best-of-family — and a second time-series dataset (ECG5000) reproduces the raw-menu ordering (24.89pp versus 3.60pp, single split, 3-vectorizer menu; §4.1); under 25-repetition repeated CV the vectorizer dominates with disjoint CIs (4.30pp versus 0.96pp; §4.1). Conti et al.'s image results are a complementary case study on 10-class MNIST under grid search, not a general law. On synthetic sphere/torus data, the combined geometric signal survives $\sigma = 0.30$ additive spatial Gaussian noise: mean accuracy across the 112 configurations at that level is 99.85%. Measured bottleneck distances (max 0.434, versus the corrected bound $2\sigma\sqrt{2 \ln n} \approx 1.82$) confirm the stability bound is conservative. A matched genus-1/genus-2 experiment with identical norm distributions shows the robustness is not an artefact of the norm confound (TDA 95.83% versus 48–58% for norm features at $\sigma = 0.30$). TDA features do not beat raw features on the datasets studied (raw-signal logistic 85.28% (r=25) vs. 82.82% TDA on ECG200 (r=25 best configuration, cubical + Persistence Image + Random Forest; the 83.0% figure is the single-split best, §4.1); raw-pixel logistic 99.65% vs. 98.0% TDA on MNIST), so the contribution is the internal TDA-pipeline comparison.

The population-level mixed model across the panel ranks the classifier first (1.81pp vs. filtration 1.71pp, vectorizer 1.18pp), and the stage×stage interactions carry more variance than any single main effect (ω^2 0.187 vs. 0.127 on ECG200), so the ordering above is a menu-conditional summary, not a population law. The core contribution

is the conditionality itself: benchmark claims about pipeline-stage importance depend on the datasets, modalities, and fixed dimensions of the study, and those conditions should be stated as explicitly as the claims. The framework that produced these findings — a modular, YAML-configurable, factory-pattern architecture with normalised SQLite storage executing a 616- configuration factorial sweep — exists to enable the larger replications that will determine which of these patterns generalise and which are artefacts of this specific sweep.

Appendix A

Full YAML Configuration

```yaml
# Expanded benchmark -- covers full framework surface area.
# Non-image filtrations silently fail on image data and vice versa;
# the runner catches and logs failures per-config.
datasets:
  - name: sphere_torus_n0
    path: data/tda/synthetic/sphere_torus_noise0_X.npy
    labels: data/tda/synthetic/sphere_torus_noise0_y.npy
    modality: point_cloud
    max_samples: 200
    subsample_points: 100
  - name: sphere_torus_n5
    path: data/tda/synthetic/sphere_torus_noise5_X.npy
    labels: data/tda/synthetic/sphere_torus_noise5_y.npy
    modality: point_cloud
    max_samples: 200
    subsample_points: 100
  - name: sphere_torus_n15
    path: data/tda/synthetic/sphere_torus_noise15_X.npy
    labels: data/tda/synthetic/sphere_torus_noise15_y.npy
    modality: point_cloud
    max_samples: 200
    subsample_points: 100
  - name: sphere_torus_n30
    path: data/tda/synthetic/sphere_torus_noise30_X.npy
    labels: data/tda/synthetic/sphere_torus_noise30_y.npy
    modality: point_cloud
    max_samples: 200
    subsample_points: 100
  - name: ecg200
    path: data/tda/ucr/ecg200_X.npy
    labels: data/tda/ucr/ecg200_y.npy
    modality: time_series
    takens_dimension: 3
```

```yaml
    takens_delay: 1
  - name: mnist_01
    path: data/tda/images/mnist_01_X.npy
    labels: data/tda/images/mnist_01_y.npy
    modality: image
    max_samples: 400
    description: "MNIST binary (0 vs 1) -- 200 per class, 28x28
       greyscale"

filtrations:
  - name: vietoris_rips
    homology_dimensions: [0, 1]
  - name: weak_alpha
    homology_dimensions: [0, 1]
  - name: sparse_rips
    homology_dimensions: [0, 1]
    epsilon: 0.3
  - name: cubical
    homology_dimensions: [0, 1]

vectorizations:
  - name: persistence_image
    sigma: 0.1
    n_bins: 20
  - name: persistence_landscape
    n_layers: 3
    n_bins: 50
  - name: betti_curve
    n_bins: 50
  - name: silhouette
    n_bins: 50
  - name: persistence_entropy
    normalize: true
  - name: amplitude
    metric: bottleneck
  - name: persistence_statistics

classifiers:
  - name: svm_rbf
  - name: random_forest
  - name: logistic
  - name: svm_linear

evaluation:
  cv_folds: 5
  scoring: accuracy
  random_seed: 42
```

```
  repetitions: 1

output:
  db_path: data/tda/expanded_results.db
  log_level: WARNING
```

Appendix B

SQLite Schema Reference

```
CREATE TABLE IF NOT EXISTS runs (
    run_id INTEGER PRIMARY KEY AUTOINCREMENT,
    dataset TEXT NOT NULL,
    filtration TEXT NOT NULL,
    vectorizer TEXT NOT NULL,
    classifier TEXT NOT NULL,
    repetition INTEGER NOT NULL,
    started_at TEXT,
    finished_at TEXT,
    wall_time_s REAL,
    peak_memory_mb REAL
);

CREATE TABLE IF NOT EXISTS fold_results (
    fold_id INTEGER PRIMARY KEY AUTOINCREMENT,
    run_id INTEGER NOT NULL REFERENCES runs(run_id),
    fold INTEGER NOT NULL,
    accuracy REAL,
    f1 REAL,
    precision REAL,
    recall REAL
);

CREATE TABLE IF NOT EXISTS run_metadata (
    run_id INTEGER PRIMARY KEY REFERENCES runs(run_id),
    pipeline_params TEXT,
    n_train INTEGER,
    n_test INTEGER,
    n_features INTEGER
);

CREATE TABLE IF NOT EXISTS config_snapshot (
    id INTEGER PRIMARY KEY CHECK (id = 1),
```

```
    yaml_text TEXT NOT NULL,
    saved_at TEXT NOT NULL
);

CREATE INDEX IF NOT EXISTS idx_runs_dataset ON runs(dataset);
CREATE INDEX IF NOT EXISTS idx_runs_filtration ON runs(filtration);
CREATE INDEX IF NOT EXISTS idx_fold_run ON fold_results(run_id);
```

Appendix C

Code Implementation

The factory classes and runner live at the repository root (`factories.py`, `runner.py`); the package is importable as `tda_benchmark` via the `pyproject.toml` mapping. Key excerpts follow.

FiltrationFactory

```python
class FiltrationFactory:
    @staticmethod
    def create(name: str, **kwargs):
        mapping = {
            "vietoris_rips": VietorisRipsPersistence,
            "sparse_rips": SparseRipsPersistence,
            "cubical": CubicalPersistence,
            "weak_alpha": WeakAlphaPersistence,
        }
        cls = mapping.get(name)
        if cls is None:
            raise ValueError(f"Unknown filtration: {name}")
        return cls(n_jobs=1, **kwargs)
```

Listing C.1: FiltrationFactory.

VectorizationFactory (auto-flatten)

```python
if name != "persistence_statistics":
    return Pipeline([
        ("vec", vec),
        ("flatten", FunctionTransformer(
            lambda x: x.reshape(x.shape[0], -1),
            validate=False)),
    ])
return vec
```

Listing C.2: VectorizationFactory auto-flatten.

Execution Loop

```
for ds, fil, vec, clf in product(datasets, filtrations,
                                 vectorizers, classifiers):
    pipeline = Pipeline([
        ("filtration",  FiltrationFactory.create(fil.name, ...)),
        ("vectorizer",  VectorizationFactory.create(vec.name, ...)),
        ("classifier",  ClassifierFactory.create(clf.name, ...)),
    ])
    scores = cross_validate(pipeline, X, y,
        cv=StratifiedKFold(n_splits=cv_folds, shuffle=True,
                   random_state=random_seed + rep)        scoring=["
                      accuracy", "f1_weighted",
                  "precision_weighted", "recall_weighted"])
```

Listing C.3: PipelineRunner execution loop.

PersistenceStatistics

```
class PersistenceStatistics(BaseEstimator, TransformerMixin):
    def transform(self, X):
        features = []
        for d in sorted(dims):
            mask = X[:, :, 2] == d
            births, deaths = X[:,:,0][mask], X[:,:,1][mask]
            lifespans = deaths - births
            for arr in [births, deaths, lifespans]:
                features.extend([np.nanmean(arr), np.nanstd(arr),
                                 np.nanmin(arr), np.nanmax(arr)])
        return np.column_stack(features)
```

Listing C.4: PersistenceStatistics transformer (simplified excerpt: the executed `_PersistenceStatistics` in `factories.py` also pads variable-length diagram lists and computes the four statistics per array *row* via axis-wise `nan`-functions, yielding one feature row per diagram; the statistics and dimensions are identical).

Appendix D

Extended Parameter Sweeps

The following sensitivity tests were conducted on reduced configuration subsets.

> **Provenance note.** The Takens sensitivity sweep below is produced by the repository's own runner worker on the full grid $d \in \{2, 3, 4\} \times \tau \in \{1, 2, 3\}$ (72 grid cells; `data/tda/takens_sweep.db`, CV seeds 43), additive and fully reproducible. At d=2 the weak-alpha filtration on 2-D embedded clouds produces essential H_1 classes (death = $+\infty$), so the six d=2 weak-alpha × persistence-image cells crash giotto-tda natively (the infinite deaths reach PersistenceImage's internal integer cast). These cells were completed by clipping the infinite deaths to the maximum finite death (flagged in the database, runs 73–78), and the table below includes these clipped cells. This clipping is the standard treatment equivalent to giotto's own landscape/Betti handling of essential classes.

Takens Embedding Sensitivity

The stage hierarchy of §4.1 was tested under alternative embedding parameters on a reduced pipeline subset (2 filtrations × 2 vectorizers × 2 classifiers):

Table D.1: Stage marginal ranges under alternative (d, τ) (reduced 8-configuration subset; range of stage-level means).

(d, τ)	Fil. Range	Vec. Range	Clf. Range
(2, 1)	0.38pp	2.12pp	7.62pp
(2, 2)	1.00pp	1.75pp	7.75pp
(2, 3)	0.12pp	2.62pp	3.88pp
(3, 1)	1.12pp	1.38pp	9.62pp
(3, 2)	1.00pp	6.00pp	8.75pp
(3, 3)	0.12pp	2.88pp	7.38pp
(4, 1)	0.50pp	5.50pp	6.50pp
(4, 2)	0.00pp	3.75pp	8.25pp
(4, 3)	1.00pp	2.00pp	7.25pp

On this reduced subset the classifier is the largest stage (3.88–9.62pp; driven by SVM-RBF collapsing to the 66.5% majority class on Persistence Entropy features) and the vectorizer second (1.38–6.00pp), while the filtration range stays at or below 1.12pp for every (d, τ) pair. The filtration-invariance of the stage analysis therefore holds under alternative embeddings; the vectorization-dominance ordering of §4.1 is measured on the full 112-configuration sweep and does not transfer to this deliberately reduced 8-configuration subset, where the vectorizer/classifier interaction dominates.

Framework Extensibility

The framework supports 7 filtrations (VR, Alpha, Sparse Rips, Čech, Cubical, Weighted Rips, Flagser), 11 vectorizers (PI, PL, Betti, Silhouette, Entropy, Amplitude, Heat Kernel, Complex Polynomials, Pairwise Distance, Number of Points, Persistence Statistics), and 4 classifiers (SVM-linear, SVM-RBF, RF, Logistic). The 616-configuration sweep reported in Chapter 4 exercises 4, 7, and 4 of these respectively. Adding a new method to the sweep requires adding one entry to the YAML configuration (Appendix A). Adding a fundamentally new component type (e.g., a learned vectorizer) requires a new factory entry in `factories.py`.

Appendix E

Use of AI Tools

AI tools were used throughout the project. This appendix records the actual workflow rather than a summary of it.

- **Benchmarking framework.** AI coding agents implemented the factory-pattern architecture, YAML loader, SQLite storage, pipeline runner, and analysis modules under my direction. I reviewed, tested, and modified the code; the smoke suite in `tests/` exercises it on every change.
- **Experiment execution.** Sweep drivers were written and launched by AI agents; every result database they produced was re-derived by independent analysis queries before any number entered this document.
- **Results paragraphs.** Agents drafted results paragraphs from database queries; I verified each number against the database it came from and edited the text before incorporation.
- **Adversarial auditing.** Multi-agent audit rounds (three-lens consensus, expanded prose lenses) were run against the draft; their findings – including errors in my favour – were verified against the databases and folded back. Audit ledgers are retained in the project records.
- **Literature synthesis and bibliography.** AI tools assisted with prior-art extraction and citation checking; every load-bearing citation was verified against its source.

All mathematical content, experimental design decisions, hypothesis choices, data interpretation, and conclusions are my own. The pre-registration file records the analysis conventions and exclusion rules fixed before the deferred sweeps ran; deviations from it are logged in its deviation section. The result databases underlying every quantitative claim ship with the archival deposit, so each number can be re-derived independently of any AI tool.

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